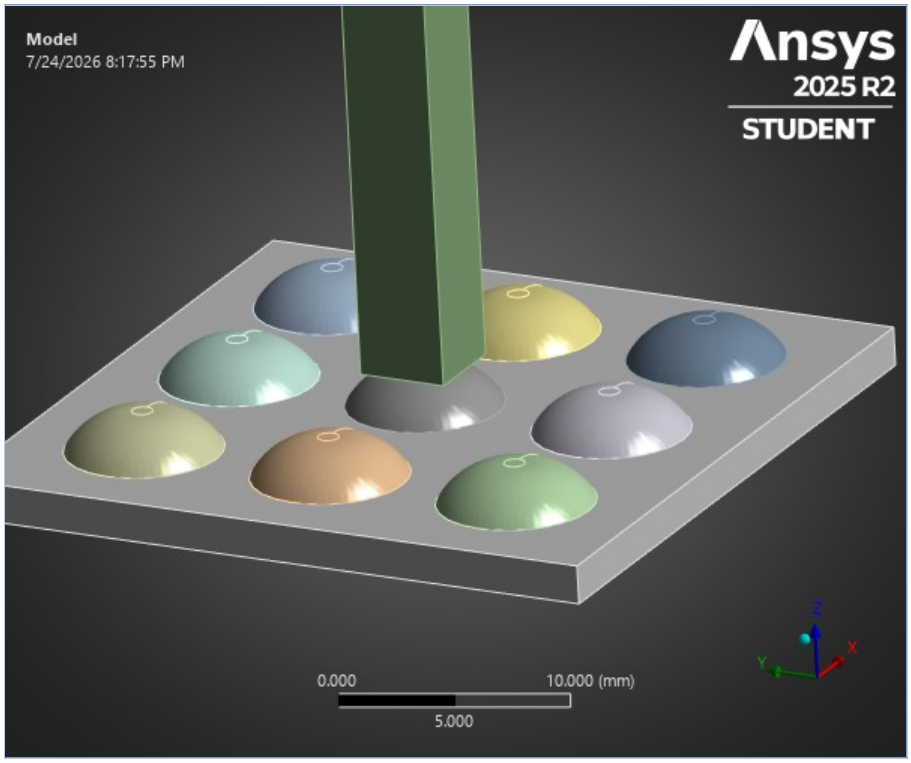
# Project*

| | |
|---|---|
| First Saved | Wednesday, July 22, 2026 |
| Last Saved | Wednesday, July 22, 2026 |
| Product Version | 2025 R2 |
| Save Project Before Solution | No |
| Save Project After Solution | No |



# Contents


- **Units**
- **Model (D4)**
  - Geometry Imports
    - Geometry Import (D3)
  - Geometry
    - Parts
  - Materials
  - Coordinate Systems
  - Connections
    - Contacts
      - Contact Regions
      - Frictionless - RUN 4|Body9 To RUN 4|Body11
  - Mesh
  - **Static Structural (D5)**
    - Analysis Settings
    - Loads
    - Solution (D6)
      - Solution Information
      - Results
      - Contact Tool
        - Status
      - Probes
- **Material Data**
  - ECOFLEX 00-45
  - Plastic, ABS (high-impact) 2


# Units

**TABLE 1**

| | |
|---|---|
| Unit System | Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius |
| Angle | Degrees |
| Rotational Velocity | rad/s |
| Temperature | Celsius |

# Model (D4)

**TABLE 2**
**Model (D4) > Geometry Imports**

| | |
|---|---|
| Object Name | *Geometry Imports* |
| State | Solved |

**TABLE 3**
**Model (D4) > Geometry Imports > Geometry Import (D3)**

| | |
|---|---|
| Object Name | *Geometry Import (D3)* |
| State | Solved |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\RUN 4.step |
| Type | Step |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |

| | |
|---|---|
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| Enclosure and Symmetry Processing | Yes |

## Geometry

**TABLE 4**
**Model (D4) > Geometry**

| Object Name | *Geometry* |
|---|---|
| State | Fully Defined |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\RUN 4.step |
| Type | Step |
| Length Unit | Millimeters |
| Element Control | Program Controlled |
| Display Style | Body Color |
| **Bounding Box** | |
| Length X | 30. mm |
| Length Y | 30. mm |
| Length Z | 41.55 mm |
| **Properties** | |
| Volume | 2343.4 mm³ |
| Mass | 2.4657e-003 kg |
| Scale Factor Value | 1. |
| **Statistics** | |
| Bodies | 11 |
| Active Bodies | 11 |
| Nodes | 7413 |
| Elements | 2829 |
| Mesh Metric | None |
| **Update Options** | |
| Assign Default Material | No |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| ID_GeometryPrefProcessPhysicsDefinition | No |
| Enclosure and Symmetry Processing | Yes |

**TABLE 5**
**Model (D4) > Geometry > Parts**

| Object Name | *RUN 4\|Body1* | *RUN 4\|Body2* | *RUN 4\|Body3* | *RUN 4\|Body4* | *RUN 4\|Body5* | *RUN 4\|Body6* | *RUN 4\|Body7* | *RUN 4\|Body8* | *RUN 4\|Body9* | *RUN 4\|Body10* | *RUN 4\|Body11* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| State | Meshed | | | | | | | | | | |
| **Graphics Properties** | | | | | | | | | | | |
| Visible | Yes | | | | | | | | | | |
| Transparency | 1 | | | | | | | | | | |
| **Definition** | | | | | | | | | | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Suppressed | No | | | | | | | | | | |
| Stiffness Behavior | Flexible | | | | | | | | | | |
| Coordinate System | Default Coordinate System | | | | | | | | | | |
| Reference Temperature | By Environment | | | | | | | | | | |
| Treatment | None | | | | | | | | | | |
| **Material** | | | | | | | | | | | |
| Assignment | ECOFLEX 00-45 | | | | | | | | | | Plastic, ABS (high-impact) 2 |
| Nonlinear Effects | Yes | | | | | | | | | | |
| Thermal Strain Effects | Yes | | | | | | | | | | |
| **Bounding Box** | | | | | | | | | | | |
| Length X | 30. mm | 7. mm | | | | | | | | | 4. mm |
| Length Y | 30. mm | 7. mm | | | | | | | | | 4. mm |
| Length Z | 1.75 mm | | | | | | | | | | 38. mm |
| **Properties** | | | | | | | | | | | |
| Volume | 1424.4 mm³ | 34.547 mm³ | | | | | | | | | 608. mm³ |
| Mass | 1.5099e-003 kg | 3.6619e-005 kg | | | | | | | | | 6.2624e-004 kg |
| Centroid X | 5.8383e-016 mm | 9. mm | -9. mm | | 2.2387e-006 mm | 9. mm | 2.2361e-006 mm | -9. mm | 2.2374e-006 mm | 9. mm | 0. mm |
| Centroid Y | 3.3172e-016 mm | 9. mm | -9. mm | 1.3304e-009 mm | 9. mm | -1.3089e-009 mm | -9. mm | 9. mm | -7.0924e-012 mm | -9. mm | 0. mm |
| Centroid Z | -0.875 mm | 0.69545 mm | | | | | | | | | 20.8 mm |
| Moment of Inertia Ip1 | 0.11686 kg·mm² | 9.7996e-005 kg·mm² | | | | | | | | | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip2 | 0.11686 kg·mm² | 9.7996e-005 kg·mm² | | | | | | | | | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip3 | 0.23295 kg·mm² | 1.8147e-004 kg·mm² | | | | | | | | | 1.67e-003 kg·mm² |
| **Statistics** | | | | | | | | | | | |
| Nodes | 1820 | 549 | 583 | 555 | 572 | 549 | 572 | 583 | 573 | 556 | 501 |
| Elements | 217 | 270 | 295 | 275 | 289 | 271 | 289 | 295 | 288 | 276 | 64 |
| Mesh Metric | None | | | | | | | | | | |

**TABLE 6**
**Model (D4) > Materials**

| Object Name | *Materials* |
|---|---|
| State | Fully Defined |
| **Statistics** | |
| Materials | 4 |
| Material Assignments | 0 |

## *Coordinate Systems*

**TABLE 7**
**Model (D4) > Coordinate Systems > Coordinate System**

| Object Name | *Global Coordinate System* |
|---|---|
| State | Fully Defined |
| **Definition** | |
| Type | Cartesian |
| Coordinate System ID | 0. |
| **Origin** | |
| Origin X | 0. mm |
| Origin Y | 0. mm |
| Origin Z | 0. mm |
| **Directional Vectors** | |
| X Axis Data | [ 1. 0. 0. ] |
| Y Axis Data | [ 0. 1. 0. ] |
| Z Axis Data | [ 0. 0. 1. ] |
| **Transfer Properties** | |
| Source | |
| Read Only | No |

## *Connections*

**TABLE 8**
**Model (D4) > Connections**

| Object Name | *Connections* |
|---|---|
| State | Fully Defined |
| **Auto Detection** | |
| Generate Automatic Connection On Refresh | Yes |
| **Transparency** | |
| Enabled | Yes |
| **Statistics** | |
| Contacts | 11 |
| Active Contacts | 11 |
| Joints | 0 |
| Active Joints | 0 |
| Beams | 0 |
| Active Beams | 0 |
| Bearings | 0 |
| Active Bearings | 0 |
| Springs | 0 |
| Active Springs | 0 |
| Body Interactions | 0 |
| Active Body Interactions | 0 |

**TABLE 9**
**Model (D4) > Connections > Contacts**

| Object Name | *Contacts* | *Contacts 2* |
|---|---|---|
| State | Fully Defined | |
| **Definition** | | |
| Connection Type | Contact | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | All Bodies | |
| **Auto Detection** | | |
| Tolerance Type | Slider | |
| Tolerance Slider | 0. | |
| Tolerance Value | 0.14846 mm | |
| Use Range | No | |
| Face/Face | Yes | |
| Face-Face Angle Tolerance | 75. ° | |
| Face Overlap Tolerance | Off | |
| Cylindrical Faces | Include | |
| Face/Edge | No | |
| Edge/Edge | No | |
| Priority | Include All | |
| Group By | Bodies | |
| Search Across | Bodies | |
| **Statistics** | | |
| Connections | 10 | 1 |
| Active Connections | 10 | 1 |

**TABLE 10**
**Model (D4) > Connections > Contacts > Contact Regions**

| Object Name | *Contact Region* | *Contact Region 2* | *Contact Region 3* | *Contact Region 4* | *Contact Region 5* | *Contact Region 6* | *Contact Region 7* | *Contact Region 8* | *Contact Region 9* | *Contact Region 10* |
|---|---|---|---|---|---|---|---|---|---|---|
| State | Fully Defined | | | | | | | | | |
| **Scope** | | | | | | | | | | |
| Scoping Method | Geometry Selection | | | | | | | | | |
| Contact | 1 Face | | | | | | | | | 2 Faces |
| Target | 1 Face | | | | | | | | | |
| Contact Bodies | RUN 4\|Body1 | | | | | | | | | RUN 4\|Body9 |
| Target Bodies | RUN 4\|Body2 | RUN 4\|Body3 | RUN 4\|Body4 | RUN 4\|Body5 | RUN 4\|Body6 | RUN 4\|Body7 | RUN 4\|Body8 | RUN 4\|Body9 | RUN 4\|Body10 | RUN 4\|Body11 |
| Protected | No | | | | | | | | | |
| **Definition** | | | | | | | | | | |
| Type | Bonded | | | | | | | | | |
| Scope Mode | Automatic | | | | | | | | | |
| Behavior | Program Controlled | | | | | | | | | |
| Trim Contact | Program Controlled | | | | | | | | | |
| Trim Tolerance | 0.14846 mm | | | | | | | | | |
| Contact APDL Name | | | | | | | | | | |

| | |
|---|---|
| Target APDL Name | |
| Suppressed | No |
| **Display** | |
| Element Normals | No |
| **Advanced** | |
| Formulation | Program Controlled |
| Small Sliding | Program Controlled |
| Detection Method | Program Controlled |
| Penetration Tolerance | Program Controlled |
| Elastic Slip Tolerance | Program Controlled |
| Normal Stiffness | Program Controlled |
| Update Stiffness | Program Controlled |
| Pinball Region | Program Controlled |
| **Geometric Modification** | |
| Contact Geometry Correction | None |
| Target Geometry Correction | None |

**TABLE 11**
**Model (D4) > Connections > Contacts 2 > Contact Regions**

| | |
|---|---|
| Object Name | *Frictionless - RUN 4\|Body9 To RUN 4\|Body11* |
| State | Fully Defined |
| **Scope** | |
| Scoping Method | Geometry Selection |
| Contact | 1 Face |
| Target | 1 Face |
| Contact Bodies | RUN 4\|Body9 |
| Target Bodies | RUN 4\|Body11 |
| Protected | No |
| **Definition** | |
| Type | Frictionless |
| Scope Mode | Manual |
| Behavior | Program Controlled |
| Trim Contact | Program Controlled |
| Contact APDL Name | |
| Target APDL Name | |
| Suppressed | No |
| **Display** | |
| Element Normals | No |
| **Advanced** | |
| Formulation | Augmented Lagrange |
| Small Sliding | Program Controlled |
| Detection Method | Program Controlled |
| Penetration Tolerance | Program Controlled |
| Normal Stiffness | Program Controlled |
| Update Stiffness | Program Controlled |
| Stabilization Damping Factor | 0. |
| Pinball Region | Program Controlled |
| Time Step Controls | None |
| **Geometric Modification** | |
| Interface Treatment | Add Offset, No Ramping |
| Offset | 0. mm |
| Contact Geometry Correction | None |
| Target Geometry Correction | None |

## *Mesh*

**TABLE 12**
**Model (D4) > Mesh**

| | |
|---|---|
| Object Name | *Mesh* |
| State | Solved |
| **Display** | |
| Display Style | Use Geometry Setting |
| **Defaults** | |
| Physics Preference | Mechanical |
| Element Order | Program Controlled |

| | |
|---|---|
| Element Size | Default |
| **Sizing** | |
| Use Adaptive Sizing | Yes |
| Resolution | Default (2) |
| Mesh Defeaturing | Yes |
| Defeature Size | Default |
| Transition | Fast |
| Span Angle Center | Coarse |
| Initial Size Seed | Assembly |
| Bounding Box Diagonal | 59.384 mm |
| Average Surface Area | 52.704 mm² |
| Minimum Edge Length | 1.061 mm |
| **Quality** | |
| Check Mesh Quality | Yes, Errors |
| Error Limits | Aggressive Mechanical |
| Target Element Quality | Default (5.e-002) |
| Smoothing | Medium |
| Mesh Metric | None |
| **Inflation** | |
| Use Automatic Inflation | None |
| Inflation Option | Smooth Transition |
| Transition Ratio | 0.272 |
| Maximum Layers | 5 |
| Growth Rate | 1.2 |
| Inflation Algorithm | Pre |
| Inflation Element Type | Wedges |
| View Advanced Options | No |
| **Advanced** | |
| Number of CPUs for Parallel Part Meshing | Program Controlled |
| Straight Sided Elements | No |
| Rigid Body Behavior | Dimensionally Reduced |
| Triangle Surface Mesher | Program Controlled |
| Topology Checking | Yes |
| Pinch Tolerance | Please Define |
| Generate Pinch on Refresh | No |
| Auto-Map Fillets | No |
| **Automatic Methods** | |
| Sheet Body Method | Prime Quad Dominant |
| Sweepable Body Method | Sweep |
| **Statistics** | |
| Nodes | 7413 |
| Elements | 2829 |
| Show Detailed Statistics | No |

## Static Structural (D5)

**TABLE 13**
**Model (D4) > Analysis**

| | |
|---|---|
| Object Name | *Static Structural (D5)* |
| State | Solved |
| **Definition** | |
| Physics Type | Structural |
| Analysis Type | Static Structural |
| Solver Target | Mechanical APDL |
| **Options** | |
| Environment Temperature | 22. °C |
| Generate Input Only | No |

**TABLE 14**
**Model (D4) > Static Structural (D5) > Analysis Settings**

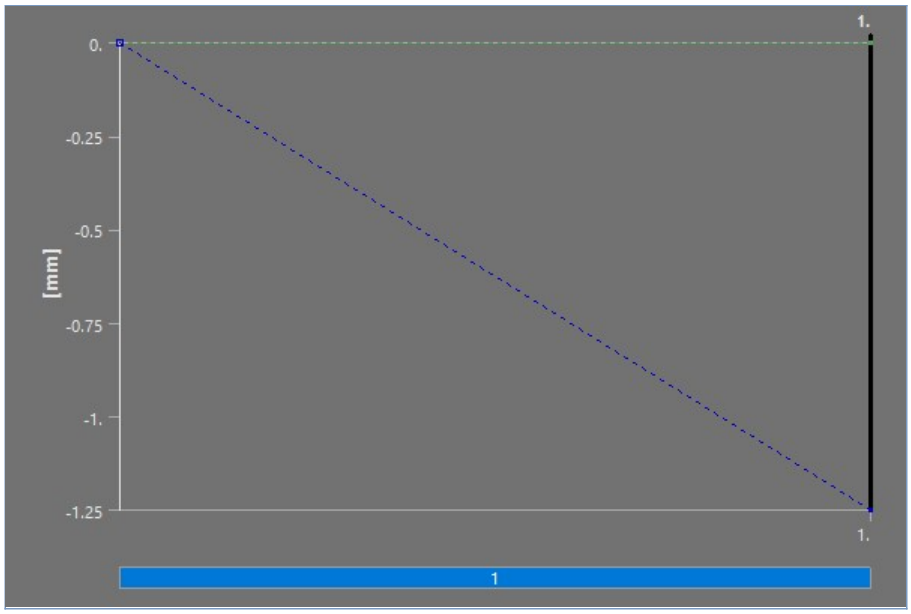
| | |
|---|---|
| Object Name | *Analysis Settings* |
| State | Fully Defined |
| **Step Controls** | |
| Number Of Steps | 1. |
| Current Step Number | 1. |
| Step End Time | 1. s |
| Auto Time Stepping | On |
| Define By | Substeps |
| Initial Substeps | 20. |
| Minimum Substeps | 5. |

| | |
|---|---|
| Maximum Substeps | 500. |
| **Solver Controls** | |
| Solver Type | Program Controlled |
| Weak Springs | On |
| Spring Stiffness | Program Controlled |
| Solver Pivot Checking | Program Controlled |
| Large Deflection | On |
| Inertia Relief | Off |
| Quasi-Static Solution | Off |
| **Rotordynamics Controls** | |
| Coriolis Effect | Off |
| **Restart Controls** | |
| Generate Restart Points | Program Controlled |
| Retain Files After Full Solve | No |
| Combine Restart Files | Program Controlled |
| **Nonlinear Controls** | |
| Newton-Raphson Option | Program Controlled |
| Force Convergence | Program Controlled |
| Moment Convergence | Program Controlled |
| Displacement Convergence | Program Controlled |
| Rotation Convergence | Program Controlled |
| Line Search | Program Controlled |
| Stabilization | Program Controlled |
| **Advanced** | |
| Inverse Option | No |
| Contact Split (DMP) | Program Controlled |
| **Output Controls** | |
| Output Selection | None |
| Stress | Yes |
| Back Stress | No |
| Strain | Yes |
| Contact Data | Yes |
| Nonlinear Data | No |
| Nodal Forces | No |
| Volume and Energy | Yes |
| Euler Angles | Yes |
| General Miscellaneous | No |
| Contact Miscellaneous | No |
| Store Results At | All Time Points |
| Result File Compression | Program Controlled |
| **Analysis Data Management** | |
| Solver Files Directory | C:\Users\DLSU\Downloads\ANSYS FOR HOLLOW RODS\MAIN PROJECT_files\dp0\SYS-9\MECH\ |
| Future Analysis | None |
| Scratch Solver Files Directory | |
| Save MAPDL db | No |
| Contact Summary | Program Controlled |
| Delete Unneeded Files | Yes |
| Nonlinear Solution | Yes |
| Solver Units | Active System |
| Solver Unit System | nmm |

**TABLE 15**
**Model (D4) > Static Structural (D5) > Loads**

| Object Name | *Fixed Support* | *Displacement* |
|---|---|---|
| State | Fully Defined | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | 1 Face | |
| **Definition** | | |
| Type | Fixed Support | Displacement |
| Suppressed | No | |
| Define By | | Components |
| Coordinate System | | Global Coordinate System |
| X Component | | 0. mm (ramped) |
| Y Component | | 0. mm (ramped) |
| Z Component | | -1.25 mm (ramped) |

**FIGURE 1**
**Model (D4) > Static Structural (D5) > Displacement**




## *Solution (D6)*

**TABLE 16**
**Model (D4) > Static Structural (D5) > Solution**

| Object Name | *Solution (D6)* |
|---|---|
| State | Solved |
| **Adaptive Mesh Refinement** | |
| Max Refinement Loops | 1. |
| Refinement Depth | 2. |
| **Information** | |
| Status | Done |
| MAPDL Elapsed Time | 34. s |
| MAPDL Memory Used | 239. MB |
| MAPDL Result File Size | 16.812 MB |
| **Post Processing** | |
| Beam Section Results | No |
| On Demand Stress/Strain | No |

**TABLE 17**
**Model (D4) > Static Structural (D5) > Solution (D6) > Solution Information**

| Object Name | *Solution Information* |
|---|---|
| State | Solved |
| **Solution Information** | |
| Solution Output | Solver Output |
| Newton-Raphson Residuals | 0 |
| Identify Element Violations | 0 |
| Update Interval | 2.5 s |
| Display Points | All |
| **FE Connection Visibility** | |
| Activate Visibility | Yes |
| Display | All FE Connectors |
| Draw Connections Attached To | All Nodes |
| Line Color | Connection Type |
| Visible on Results | No |
| Line Thickness | Single |
| Display Type | Lines |

**TABLE 18**
**Model (D4) > Static Structural (D5) > Solution (D6) > Results**

| Object Name | *Total Deformation* | *Directional Deformation* | *Directional Deformation 2* | *Directional Deformation 3* | *Equivalent Elastic Strain* | *Maximum Principal Elastic Strain* | *Maximum Shear Elastic Strain* | *Elastic Strain Intensity* | *Equivalent Stress* | *Maximum Principal Stress* | *Maximum Shear Stress* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| State | Solved | | | | | | | | | | |
| **Scope** | | | | | | | | | | | |
| Scoping | Geometry Selection | | | | | | | | | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | | | | | | | | | | | |
| Geometry | 2 Bodies | 5 Faces | | | 2 Bodies | | | | | | |
| **Definition** | | | | | | | | | | | |
| Type | Total Deformation | Directional Deformation | | | Equivalent Elastic Strain | Maximum Principal Elastic Strain | Maximum Shear Elastic Strain | Elastic Strain Intensity | Equivalent (von-Mises) Stress | Maximum Principal Stress | Maximum Shear Stress |
| By | Time | | | | | | | | | | |
| Display Time | Last | | | | | | | | | | |
| Separate Data by Entity | No | | | | | | | | | | |
| Calculate Time History | Yes | | | | | | | | | | |
| Identifier | | | | | | | | | | | |
| Suppressed | No | | | | | | | | | | |
| Orientation | | X Axis | Y Axis | Z Axis | | | | | | | |
| Coordinate System | | Global Coordinate System | | | | | | | | | |
| **Results** | | | | | | | | | | | |
| Minimum | 0. mm | -0.4575 mm | -0.42261 mm | -1.2669 mm | 2.4325e-005 mm/mm | 1.7231e-005 mm/mm | 4.207e-005 mm/mm | | 3.4518e-006 MPa | -4.6236e-002 MPa | 1.9899e-006 MPa |
| Maximum | 1.2875 mm | 0.42382 mm | 0.40017 mm | 9.4296e-002 mm | 0.54173 mm/mm | 0.30974 mm/mm | 0.83428 mm/mm | | 6.8249e-002 MPa | 3.609e-002 MPa | 3.8192e-002 MPa |
| Average | 0.23361 mm | -1.2109e-003 mm | -1.9415e-003 mm | -0.20315 mm | 6.5235e-002 mm/mm | 3.6778e-002 mm/mm | 8.9685e-002 mm/mm | | 7.4679e-003 MPa | 6.1689e-004 MPa | 4.1741e-003 MPa |
| Minimum Occurs On | RUN 4\|Body1 | | | RUN 4\|Body9 | RUN 4\|Body1 | | | | | | |
| Maximum Occurs On | RUN 4\|Body9 | | | RUN 4\|Body1 | RUN 4\|Body9 | | | | | | |
| **Minimum Value Over Time** | | | | | | | | | | | |
| Minimum | 0. mm | -0.4575 mm | -0.42261 mm | -1.2669 mm | 6.7032e-007 mm/mm | 4.5648e-007 mm/mm | 1.1574e-006 mm/mm | | 9.5118e-008 MPa | -4.6236e-002 MPa | 5.4745e-008 MPa |
| Maximum | 0. mm | -1.1692e-002 mm | -1.1814e-002 mm | -6.5307e-002 mm | 2.4325e-005 mm/mm | 1.7231e-005 mm/mm | 4.207e-005 mm/mm | | 3.4518e-006 MPa | -2.2654e-003 MPa | 1.9899e-006 MPa |
| **Maximum Value Over Time** | | | | | | | | | | | |
| Minimum | 6.5429e-002 mm | 1.1834e-002 mm | 1.1447e-002 mm | 2.104e-003 mm | 2.8376e-002 mm/mm | 2.0333e-002 mm/mm | 4.6515e-002 mm/mm | | 3.8658e-003 MPa | 1.3558e-003 MPa | 2.1888e-003 MPa |
| Maximum | 1.2875 mm | 0.42382 mm | 0.40017 mm | 9.4296e-002 mm | 0.54173 mm/mm | 0.30974 mm/mm | 0.83428 mm/mm | | 6.8249e-002 MPa | 3.609e-002 MPa | 3.8192e-002 MPa |
| **Information** | | | | | | | | | | | |
| Time | 1. s | | | | | | | | | | |
| Load Step | 1 | | | | | | | | | | |
| Substep | 10 | | | | | | | | | | |
| Iteration Number | 30 | | | | | | | | | | |
| **Integration Point Results** | | | | | | | | | | | |
| Display Option | | | | | Averaged | | | | | | |
| Average Across Bodies | | | | | No | | | | | | |

**FIGURE 2**
**Model (D4) > Static Structural (D5) > Solution (D6) > Total Deformation**

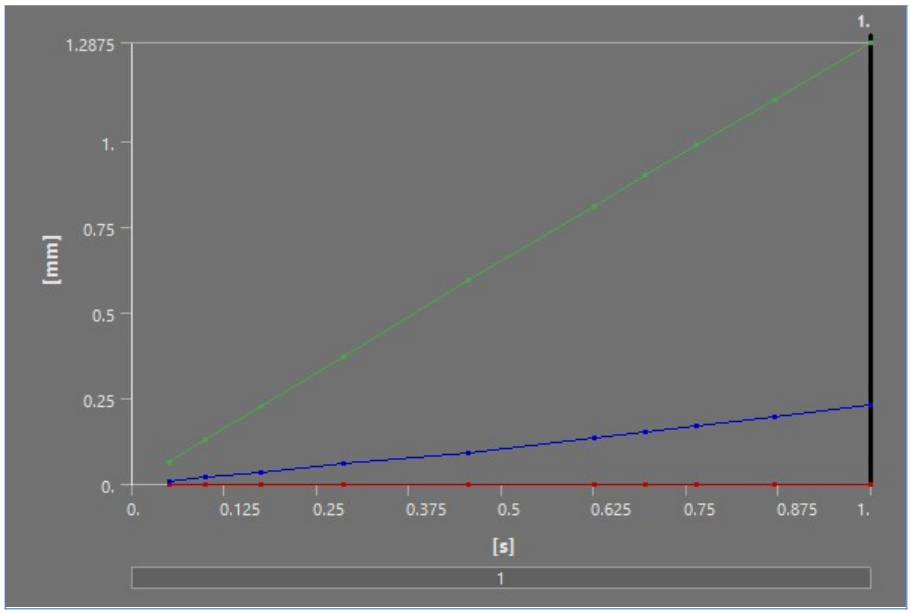



**TABLE 19**
**Model (D4) > Static Structural (D5) > Solution (D6) > Total Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | 0. | 6.5429e-002 | 1.0435e-002 |
| 0.1 | | 0.13026 | 2.0827e-002 |
| 0.175 | | 0.22754 | 3.6315e-002 |
| 0.2875 | | 0.37439 | 5.9257e-002 |
| 0.45625 | | 0.59518 | 9.3879e-002 |
| 0.625 | | 0.81205 | 0.13498 |
| 0.695 | | 0.90066 | 0.15311 |
| 0.765 | | 0.9893 | 0.17145 |
| 0.87 | | 1.1224 | 0.19912 |
| 1. | | 1.2875 | 0.23361 |

**FIGURE 3**
**Model (D4) > Static Structural (D5) > Solution (D6) > Total Deformation > Figure**

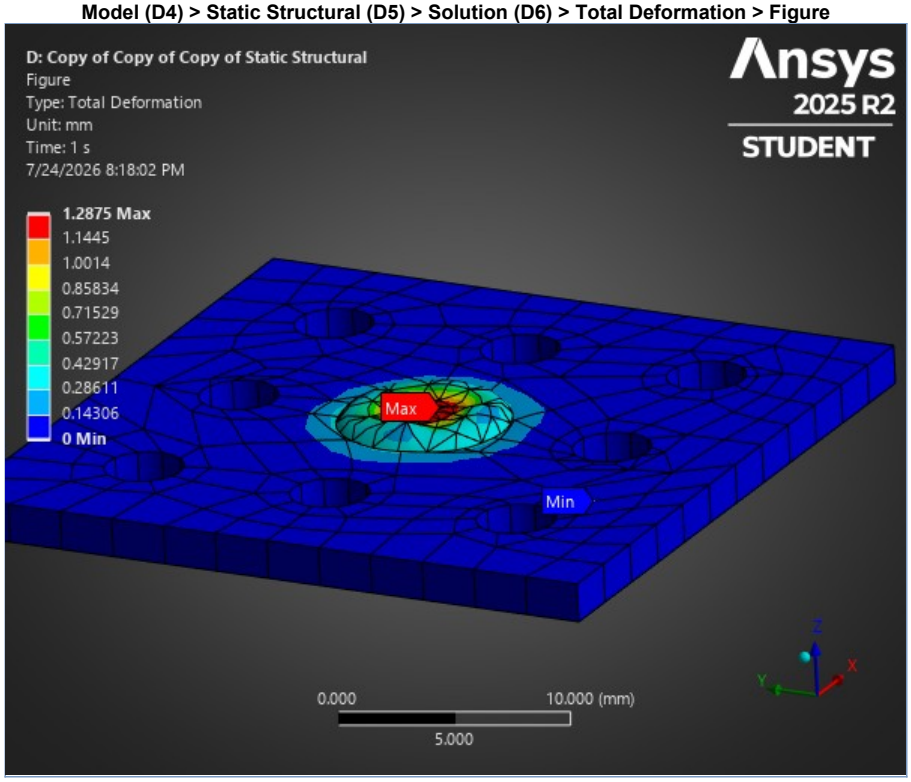


**FIGURE 4**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation**

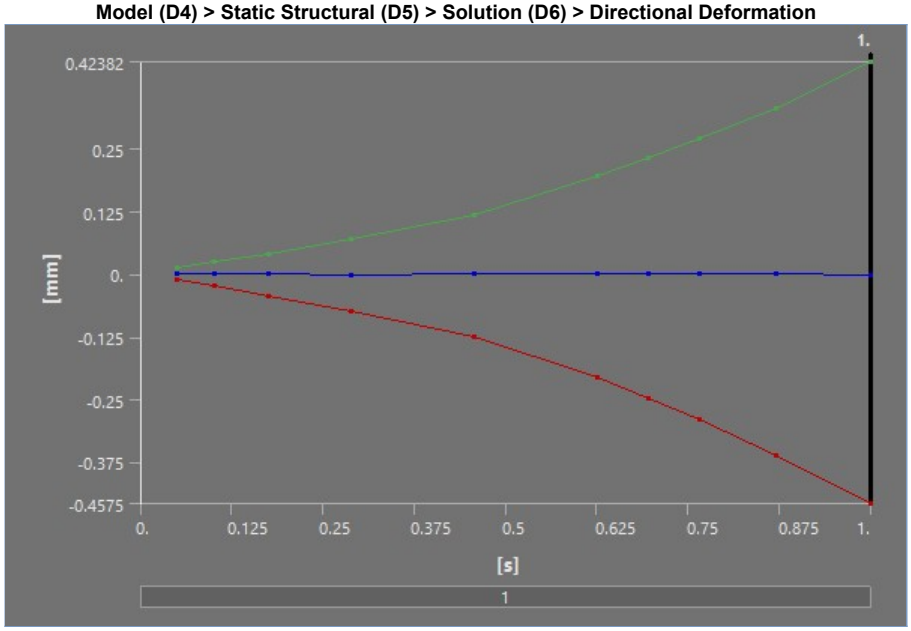



**TABLE 20**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -1.1692e-002 | 1.1834e-002 | -4.2836e-005 |
| 0.1 | -2.3742e-002 | 2.376e-002 | -9.6217e-005 |
| 0.175 | -4.2473e-002 | 4.1608e-002 | -1.9633e-004 |
| 0.2875 | -7.4411e-002 | 7.0074e-002 | -3.909e-004 |
| 0.45625 | -0.12508 | 0.11845 | -9.2774e-005 |
| 0.625 | -0.20614 | 0.19468 | 7.3904e-004 |
| 0.695 | -0.24701 | 0.2319 | 4.2619e-004 |
| 0.765 | -0.2907 | 0.27095 | 1.8803e-004 |
| 0.87 | -0.36136 | 0.33206 | -3.4154e-004 |
| 1. | -0.4575 | 0.42382 | -1.2109e-003 |

**FIGURE 5**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation > Figure**

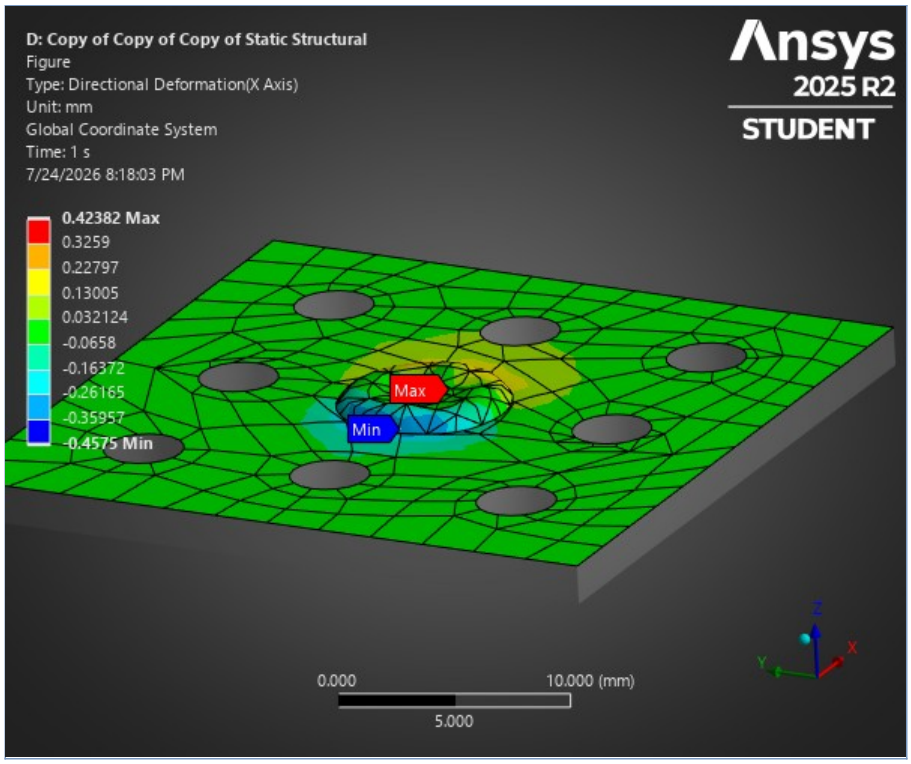



**FIGURE 6**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 2**

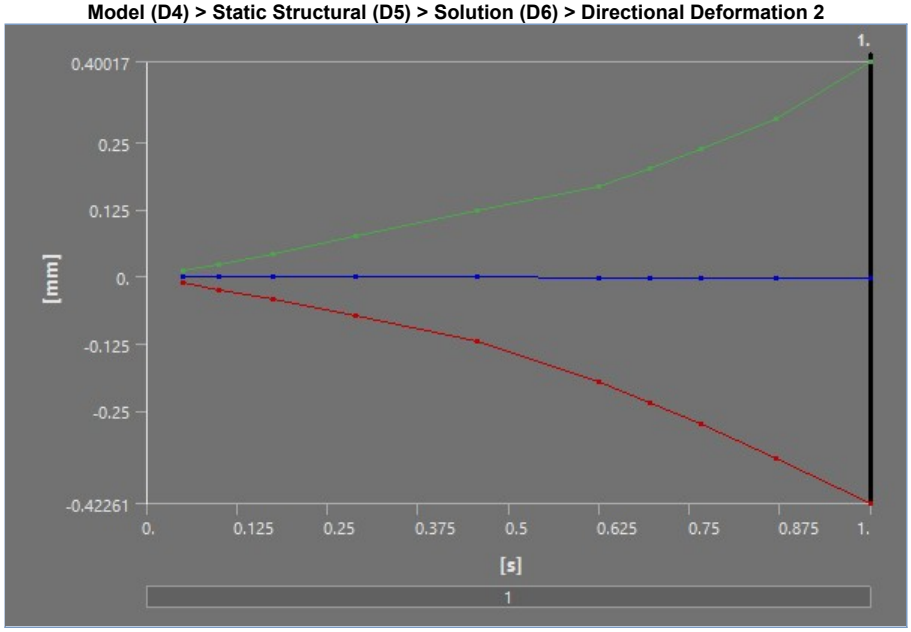



**TABLE 21**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 2**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -1.1814e-002 | 1.1447e-002 | -1.0537e-004 |
| 0.1 | -2.3873e-002 | 2.348e-002 | -2.1587e-004 |
| 0.175 | -4.2367e-002 | 4.2674e-002 | -3.9386e-004 |
| 0.2875 | -7.296e-002 | 7.4349e-002 | -6.9578e-004 |
| 0.45625 | -0.12007 | 0.12285 | -1.2461e-003 |
| 0.625 | -0.19624 | 0.169 | -1.5992e-003 |
| 0.695 | -0.23479 | 0.20256 | -1.6197e-003 |
| 0.765 | -0.27531 | 0.23779 | -1.5863e-003 |
| 0.87 | -0.33908 | 0.29272 | -1.6927e-003 |
| 1. | -0.42261 | 0.40017 | -1.9415e-003 |

**FIGURE 7**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 2 > Figure**

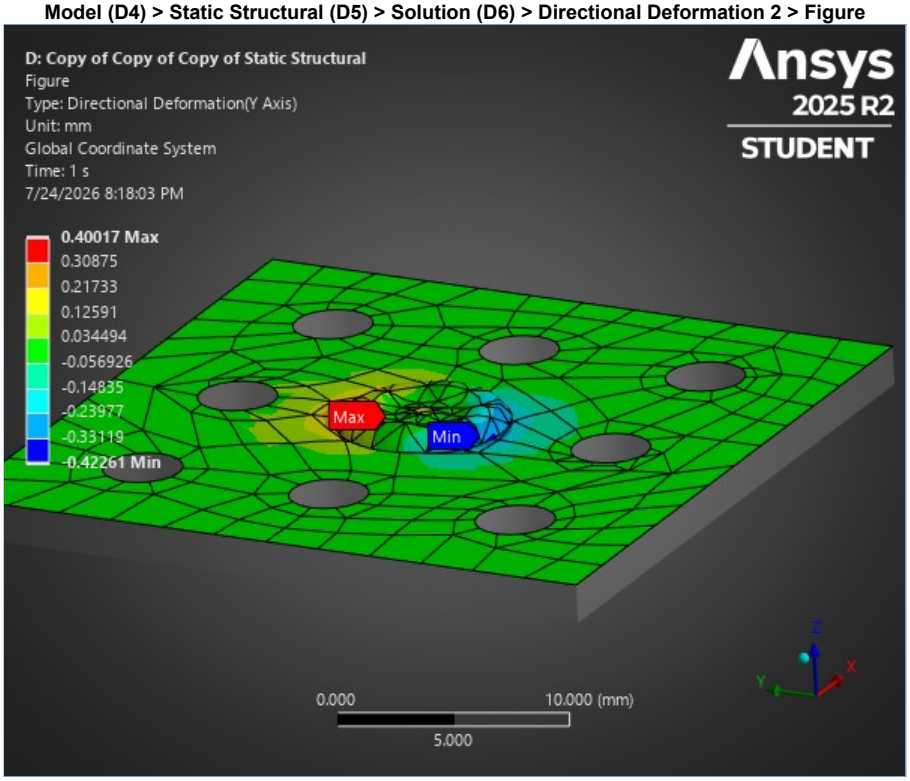



**FIGURE 8**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 3**

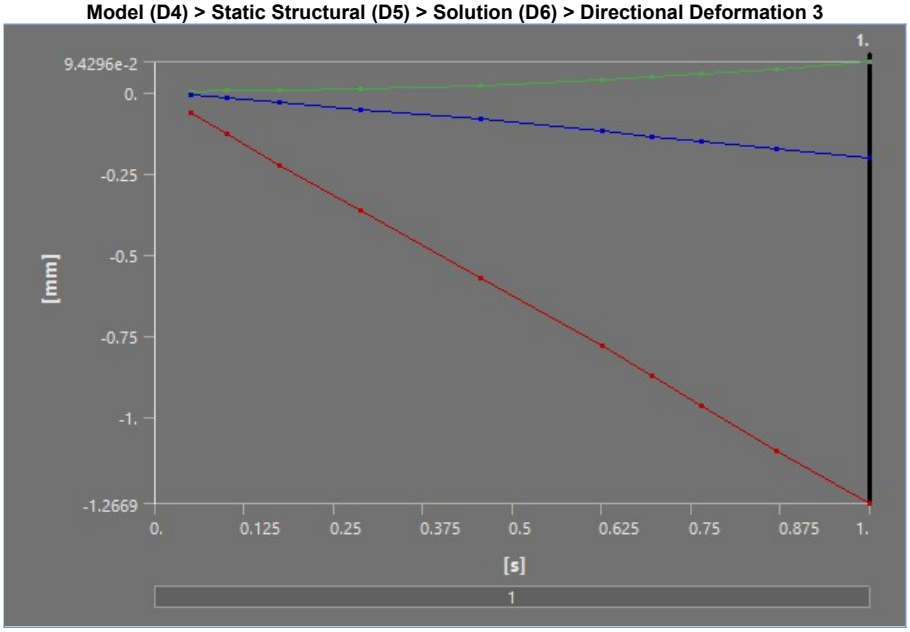



**TABLE 22**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 3**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -6.5307e-002 | 2.104e-003 | -9.4019e-003 |
| 0.1 | -0.12972 | 4.2443e-003 | -1.8699e-002 |
| 0.175 | -0.22451 | 7.6567e-003 | -3.2428e-002 |
| 0.2875 | -0.36223 | 1.3085e-002 | -5.2444e-002 |
| 0.45625 | -0.57031 | 2.1907e-002 | -8.2293e-002 |
| 0.625 | -0.78125 | 3.7333e-002 | -0.11937 |
| 0.695 | -0.87488 | 4.6355e-002 | -0.13554 |
| 0.765 | -0.96778 | 5.5868e-002 | -0.15161 |
| 0.87 | -1.1044 | 7.1817e-002 | -0.17516 |

| | | | |
|---|---|---|---|
| 1. | -1.2669 | 9.4296e-002 | -0.20315 |

**FIGURE 9**
**Model (D4) > Static Structural (D5) > Solution (D6) > Directional Deformation 3 > Figure**

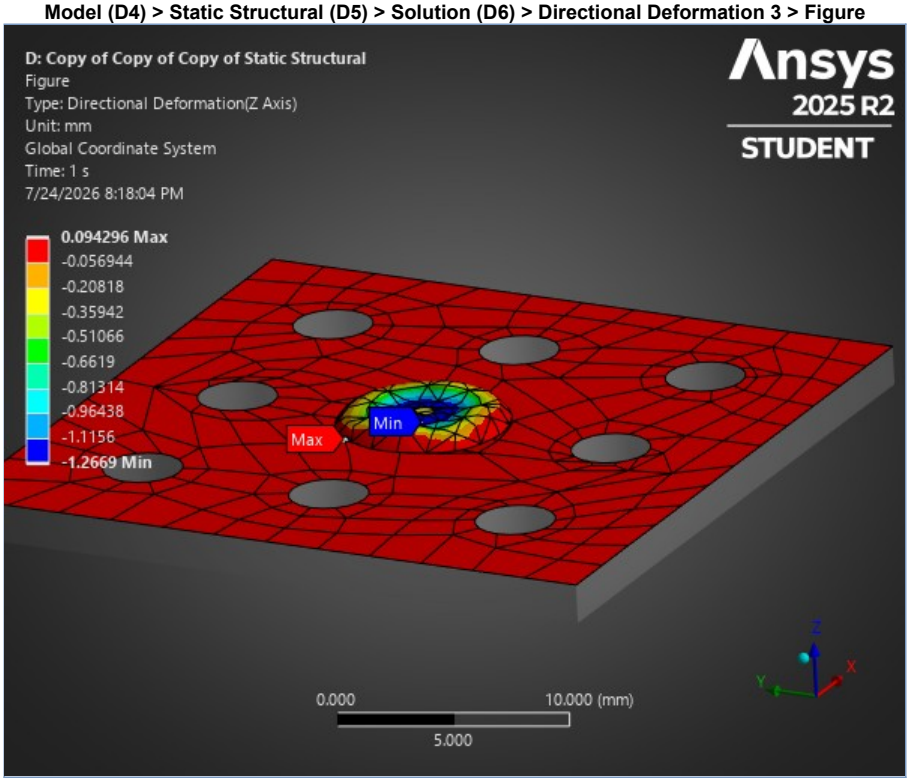



**FIGURE 10**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Elastic Strain**

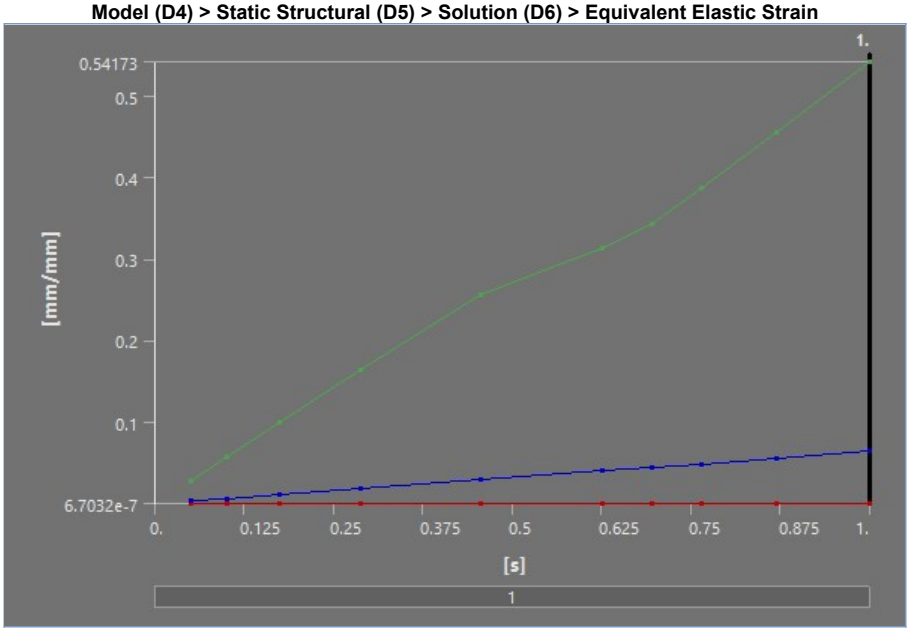



**TABLE 23**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 6.7032e-007 | 2.8376e-002 | 3.146e-003 |
| 0.1 | 1.3501e-006 | 5.6952e-002 | 6.2921e-003 |
| 0.175 | 2.384e-006 | 0.10003 | 1.1063e-002 |
| 0.2875 | 3.9462e-006 | 0.16473 | 1.8525e-002 |
| 0.45625 | 6.5128e-006 | 0.25528 | 3.0354e-002 |
| 0.625 | 1.1055e-005 | 0.31396 | 3.9747e-002 |
| 0.695 | 1.332e-005 | 0.34233 | 4.3594e-002 |

| | | | |
|---|---|---|---|
| 0.765 | 1.5704e-005 | 0.38716 | 4.7822e-002 |
| 0.87 | 1.9437e-005 | 0.4559 | 5.49e-002 |
| 1. | 2.4325e-005 | 0.54173 | 6.5235e-002 |

**FIGURE 11**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Elastic Strain > Figure**

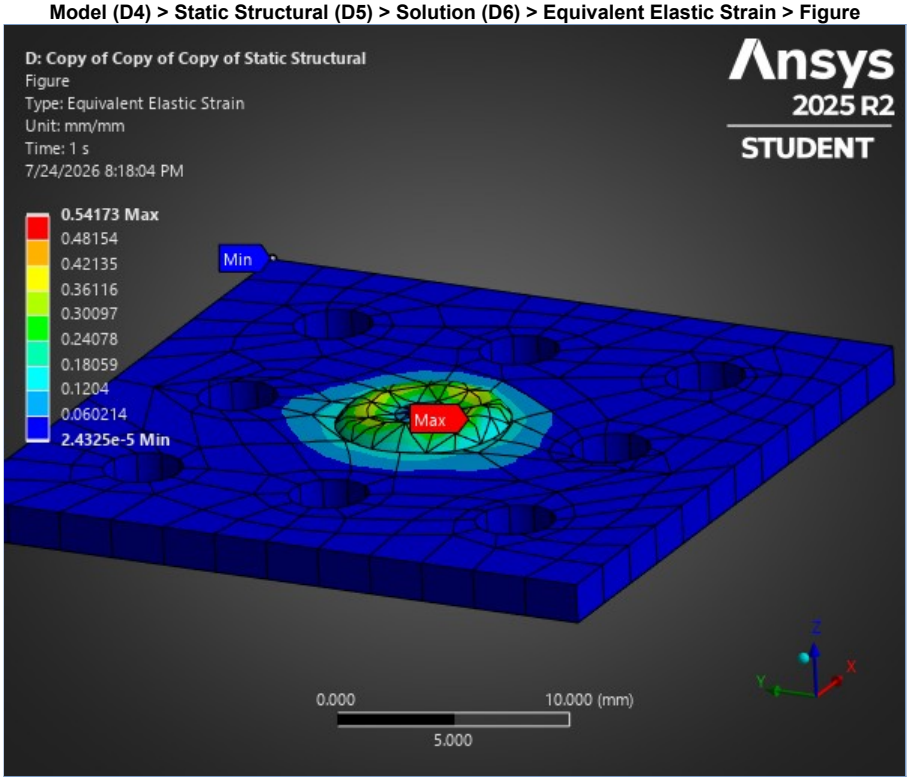



**FIGURE 12**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Elastic Strain**

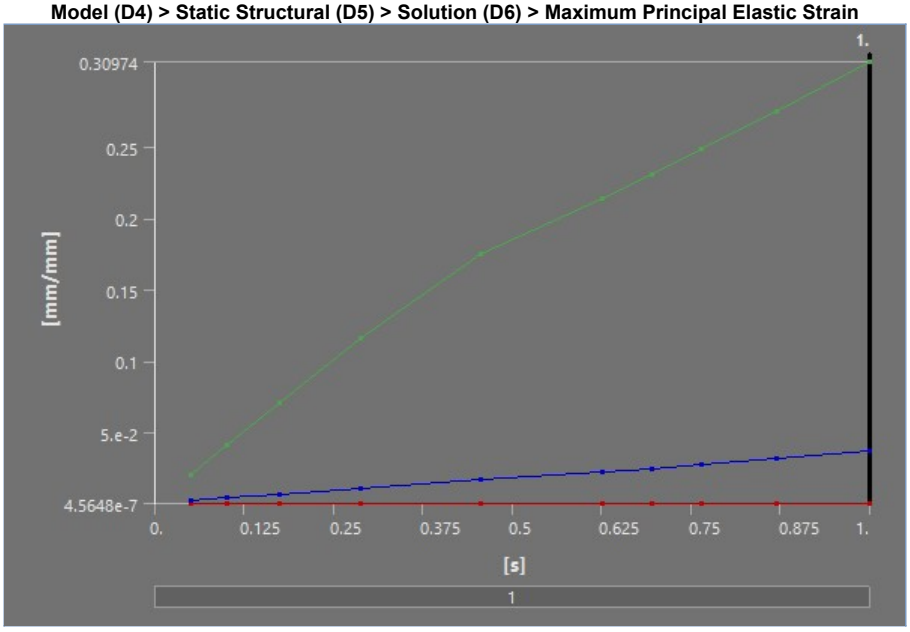



**TABLE 24**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 4.5648e-007 | 2.0333e-002 | 1.8651e-003 |
| 0.1 | 9.2036e-007 | 4.0656e-002 | 3.6957e-003 |
| 0.175 | 1.6282e-006 | 7.0944e-002 | 6.4087e-003 |
| 0.2875 | 2.7041e-006 | 0.11543 | 1.0499e-002 |
| 0.45625 | 4.5199e-006 | 0.17483 | 1.6793e-002 |

| | | | |
|---|---|---|---|
| 0.625 | 7.6839e-006 | 0.21366 | 2.2334e-002 |
| 0.695 | 9.2538e-006 | 0.23109 | 2.4707e-002 |
| 0.765 | 1.0918e-005 | 0.24899 | 2.7254e-002 |
| 0.87 | 1.3586e-005 | 0.27497 | 3.1329e-002 |
| 1. | 1.7231e-005 | 0.30974 | 3.6778e-002 |

**FIGURE 13**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Elastic Strain > Figure**

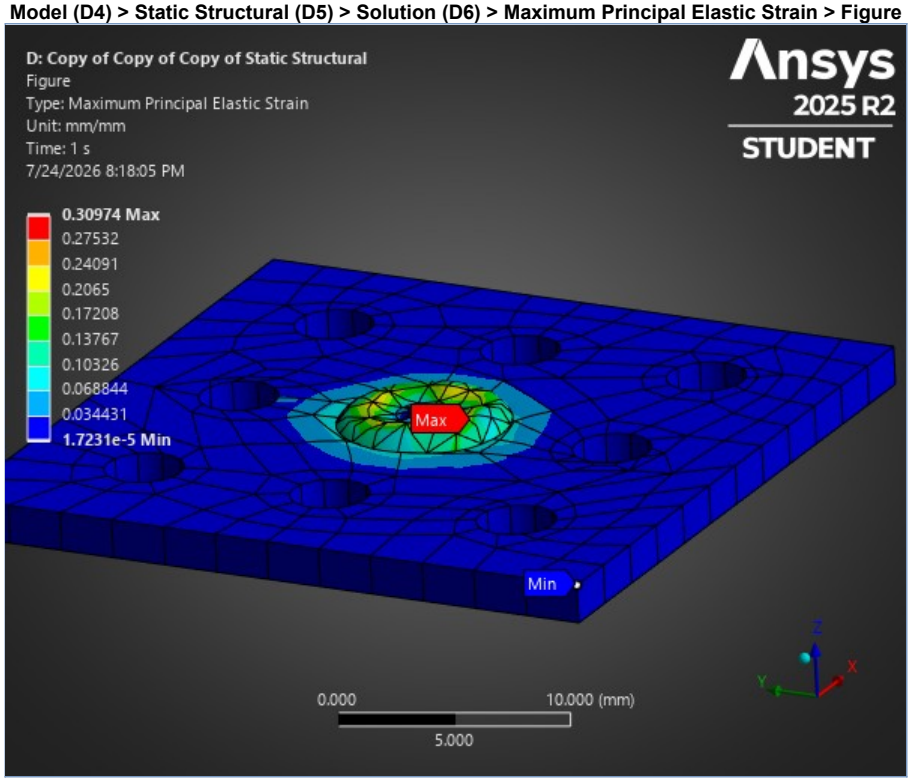



**FIGURE 14**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Elastic Strain**

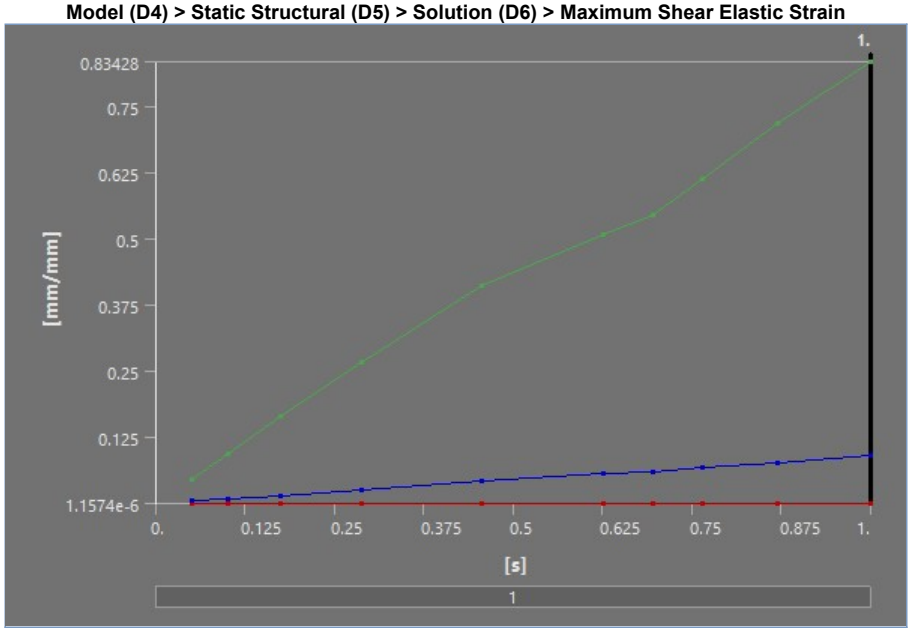



**TABLE 25**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 1.1574e-006 | 4.6515e-002 | 4.4246e-003 |
| 0.1 | 2.331e-006 | 9.3222e-002 | 8.8295e-003 |
| 0.175 | 4.1157e-006 | 0.16333 | 1.5471e-002 |

| | | | |
|---|---|---|---|
| 0.2875 | 6.8113e-006 | 0.26773 | 2.5736e-002 |
| 0.45625 | 1.1242e-005 | 0.41068 | 4.1854e-002 |
| 0.625 | 1.9102e-005 | 0.50814 | 5.5455e-002 |
| 0.695 | 2.3022e-005 | 0.54465 | 6.1002e-002 |
| 0.765 | 2.7148e-005 | 0.61435 | 6.6938e-002 |
| 0.87 | 3.3608e-005 | 0.71733 | 7.6516e-002 |
| 1. | 4.207e-005 | 0.83428 | 8.9685e-002 |

**FIGURE 15**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Elastic Strain > Figure**

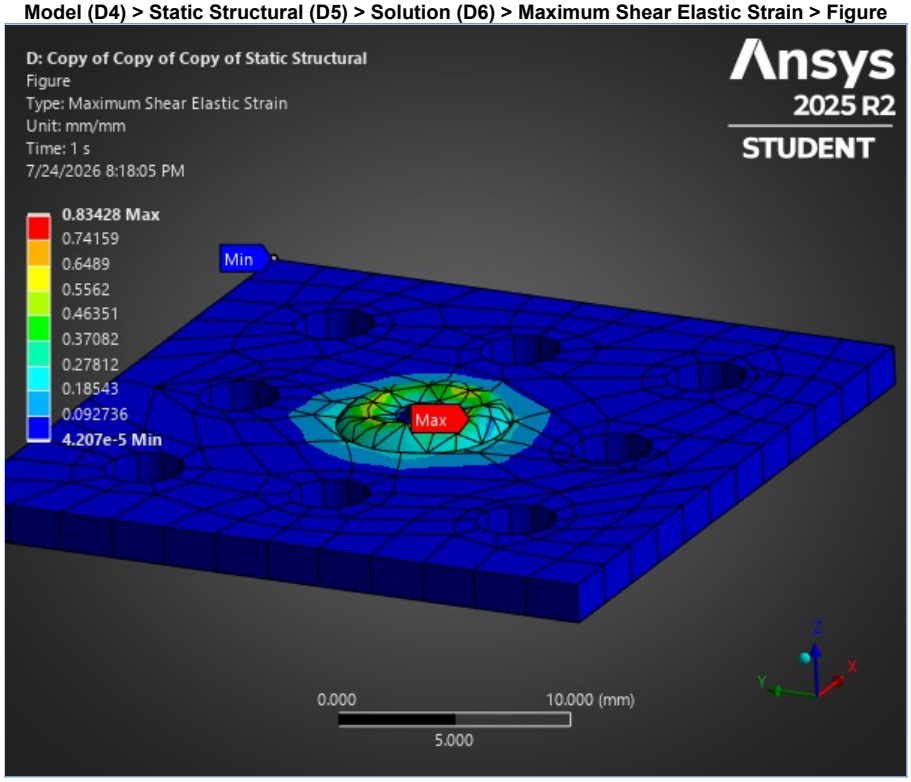



**FIGURE 16**
**Model (D4) > Static Structural (D5) > Solution (D6) > Elastic Strain Intensity**

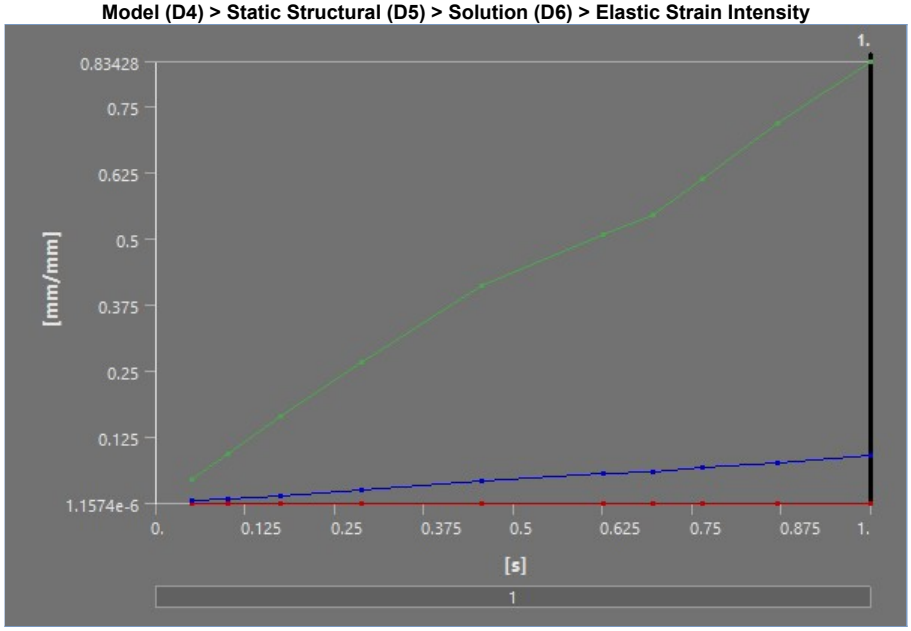



**TABLE 26**
**Model (D4) > Static Structural (D5) > Solution (D6) > Elastic Strain Intensity**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 1.1574e-006 | 4.6515e-002 | 4.4246e-003 |

| | | | |
|---|---|---|---|
| 0.1 | 2.331e-006 | 9.3222e-002 | 8.8295e-003 |
| 0.175 | 4.1157e-006 | 0.16333 | 1.5471e-002 |
| 0.2875 | 6.8113e-006 | 0.26773 | 2.5736e-002 |
| 0.45625 | 1.1242e-005 | 0.41068 | 4.1854e-002 |
| 0.625 | 1.9102e-005 | 0.50814 | 5.5455e-002 |
| 0.695 | 2.3022e-005 | 0.54465 | 6.1002e-002 |
| 0.765 | 2.7148e-005 | 0.61435 | 6.6938e-002 |
| 0.87 | 3.3608e-005 | 0.71733 | 7.6516e-002 |
| 1. | 4.207e-005 | 0.83428 | 8.9685e-002 |

**FIGURE 17**
**Model (D4) > Static Structural (D5) > Solution (D6) > Elastic Strain Intensity > Figure**

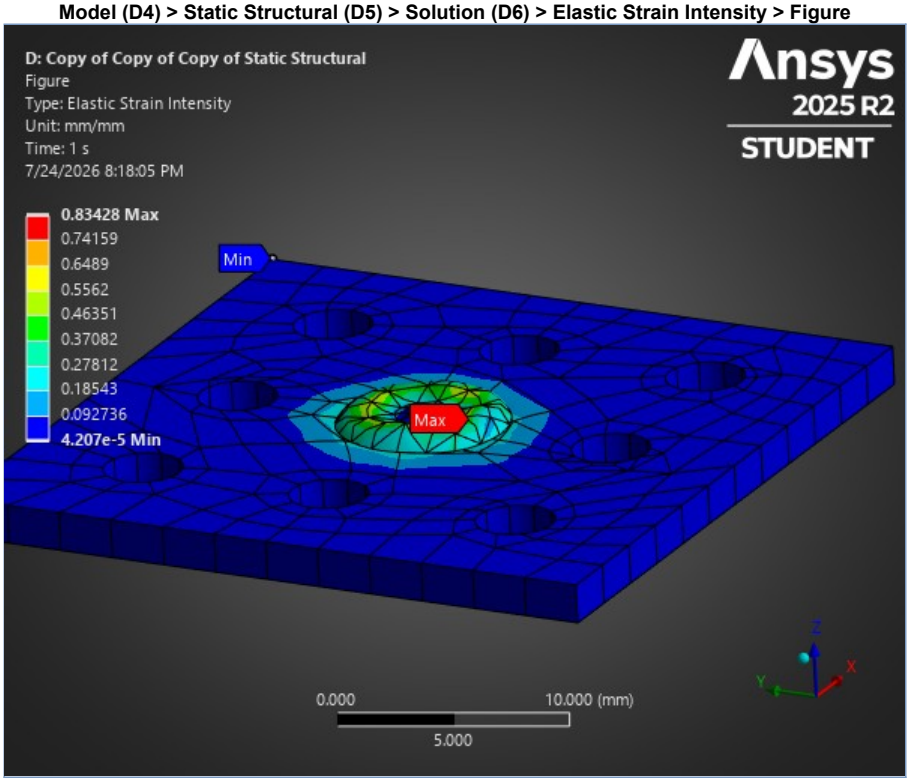



**FIGURE 18**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Stress**

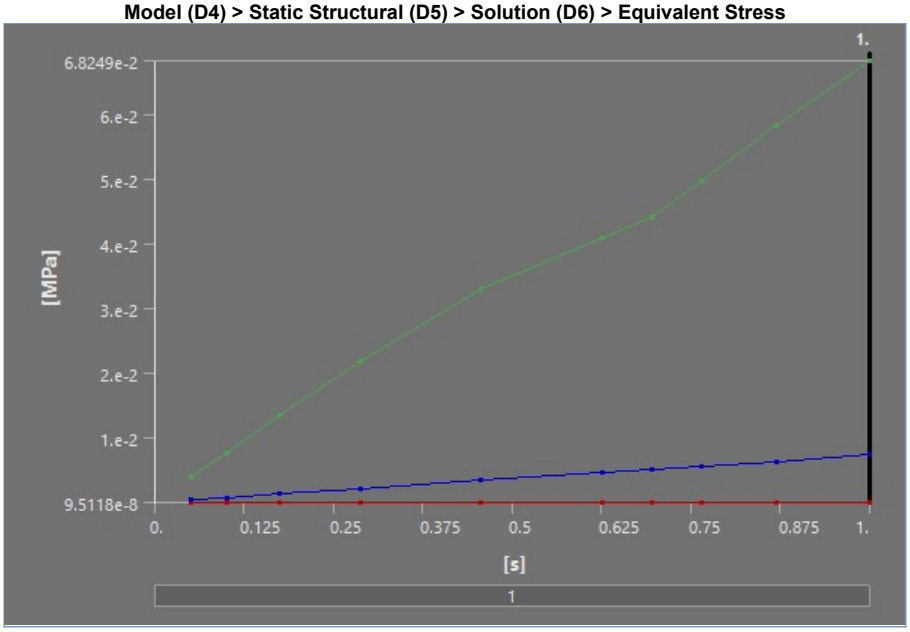



**TABLE 27**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 9.5118e-008 | 3.8658e-003 | 3.7929e-004 |
| 0.1 | 1.9158e-007 | 7.6998e-003 | 7.5371e-004 |
| 0.175 | 3.3829e-007 | 1.3383e-002 | 1.3134e-003 |
| 0.2875 | 5.5996e-007 | 2.1735e-002 | 2.1702e-003 |
| 0.45625 | 9.2417e-007 | 3.3038e-002 | 3.5022e-003 |
| 0.625 | 1.5687e-006 | 4.0828e-002 | 4.6127e-003 |
| 0.695 | 1.8901e-006 | 4.4065e-002 | 5.062e-003 |
| 0.765 | 2.2284e-006 | 4.9714e-002 | 5.548e-003 |
| 0.87 | 2.7581e-006 | 5.8222e-002 | 6.3449e-003 |
| 1. | 3.4518e-006 | 6.8249e-002 | 7.4679e-003 |

**FIGURE 19**
**Model (D4) > Static Structural (D5) > Solution (D6) > Equivalent Stress > Figure**

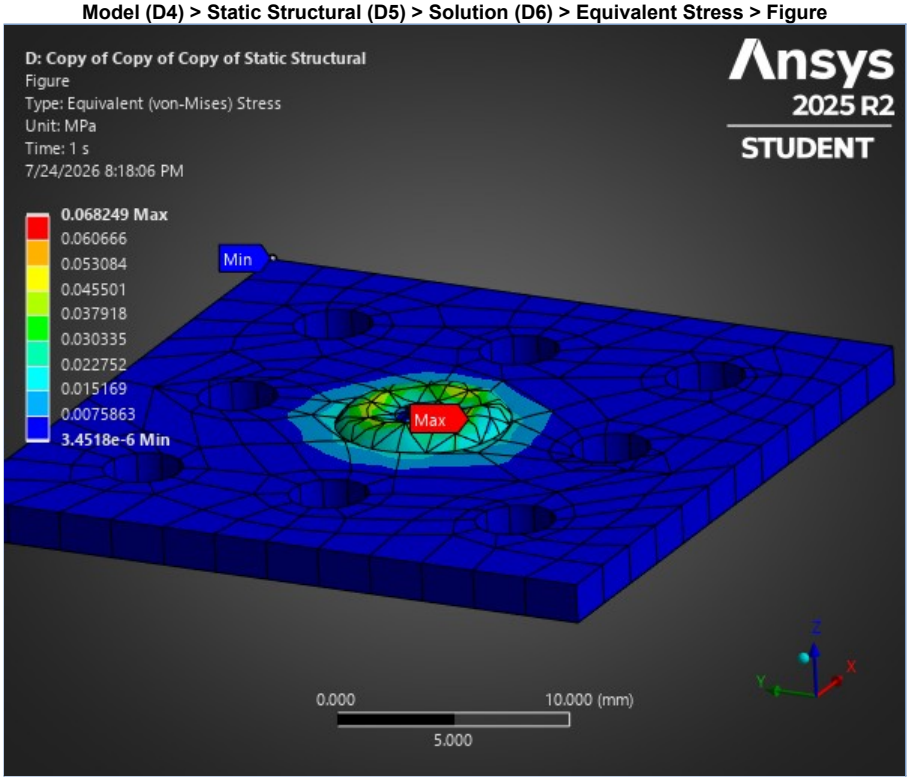



**FIGURE 20**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Stress**

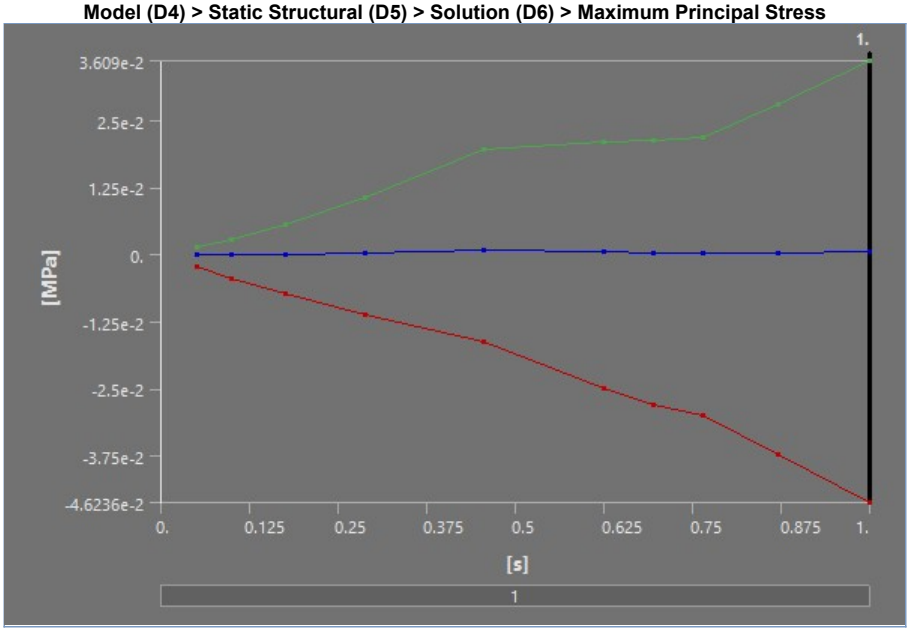


**TABLE 28**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | -2.2654e-003 | 1.3558e-003 | -8.0136e-006 |
| 0.1 | -4.4119e-003 | 2.9061e-003 | 4.6654e-006 |
| 0.175 | -7.3678e-003 | 5.6047e-003 | 6.7214e-005 |
| 0.2875 | -1.1191e-002 | 1.0719e-002 | 2.8354e-004 |
| 0.45625 | -1.6173e-002 | 1.957e-002 | 7.5072e-004 |
| 0.625 | -2.4907e-002 | 2.1068e-002 | 4.6681e-004 |
| 0.695 | -2.8035e-002 | 2.1228e-002 | 3.0368e-004 |
| 0.765 | -3.0013e-002 | 2.1874e-002 | 1.8941e-004 |
| 0.87 | -3.7257e-002 | 2.7991e-002 | 1.9367e-004 |
| 1. | -4.6236e-002 | 3.609e-002 | 6.1689e-004 |

**FIGURE 21**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Principal Stress > Figure**

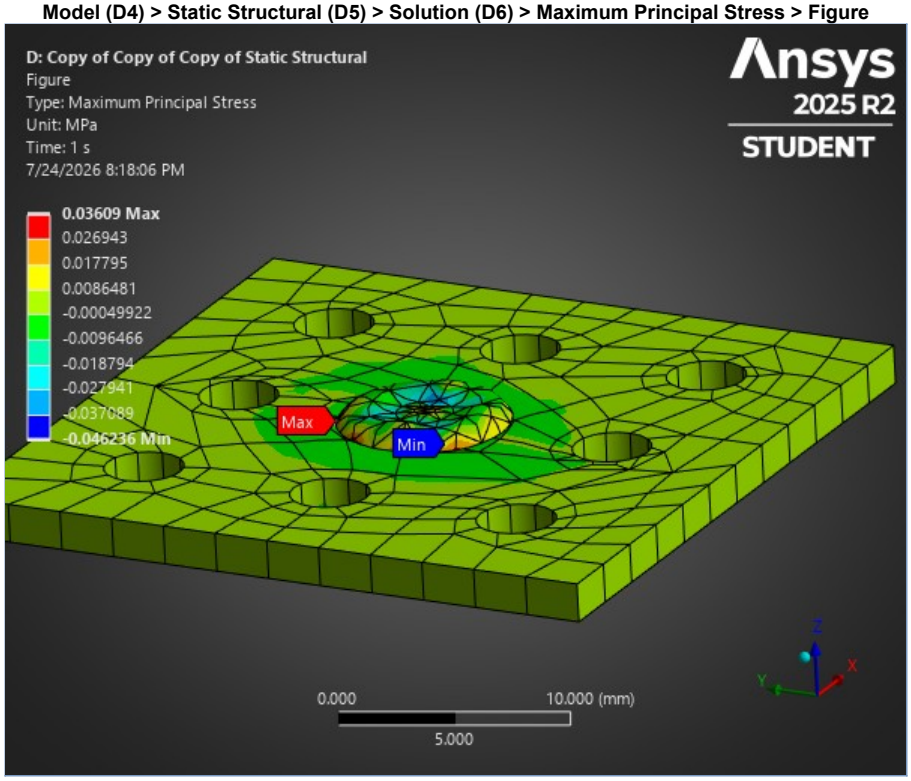



**FIGURE 22**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Stress**

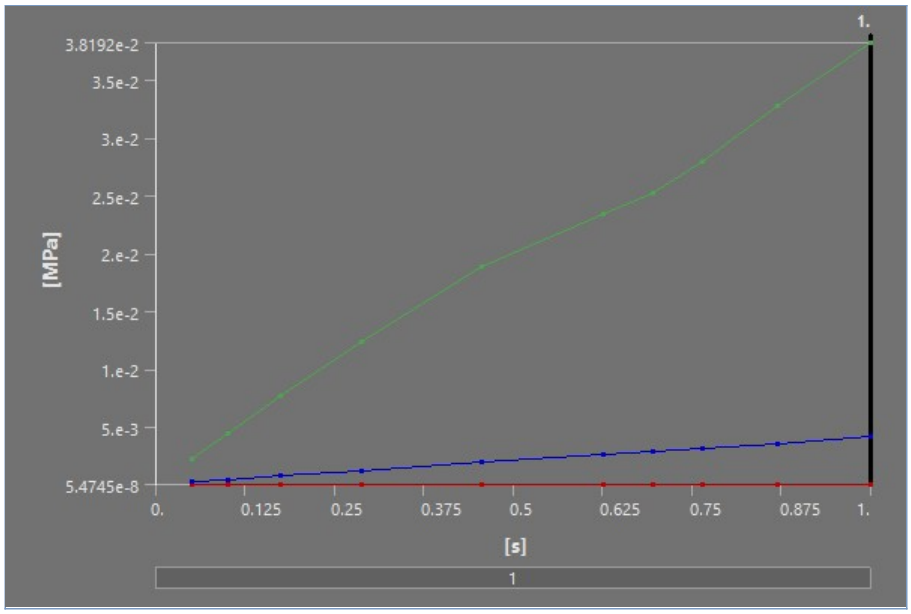



**TABLE 29**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 5.4745e-008 | 2.1888e-003 | 2.0852e-004 |
| 0.1 | 1.1026e-007 | 4.3665e-003 | 4.1464e-004 |
| 0.175 | 1.9467e-007 | 7.6055e-003 | 7.2295e-004 |
| 0.2875 | 3.2218e-007 | 1.2385e-002 | 1.1945e-003 |
| 0.45625 | 5.3176e-007 | 1.8828e-002 | 1.9263e-003 |
| 0.625 | 9.0351e-007 | 2.3339e-002 | 2.5518e-003 |
| 0.695 | 1.0889e-006 | 2.5158e-002 | 2.8107e-003 |
| 0.765 | 1.2841e-006 | 2.798e-002 | 3.0894e-003 |
| 0.87 | 1.5897e-006 | 3.276e-002 | 3.5435e-003 |
| 1. | 1.9899e-006 | 3.8192e-002 | 4.1741e-003 |

**FIGURE 23**
**Model (D4) > Static Structural (D5) > Solution (D6) > Maximum Shear Stress > Figure**

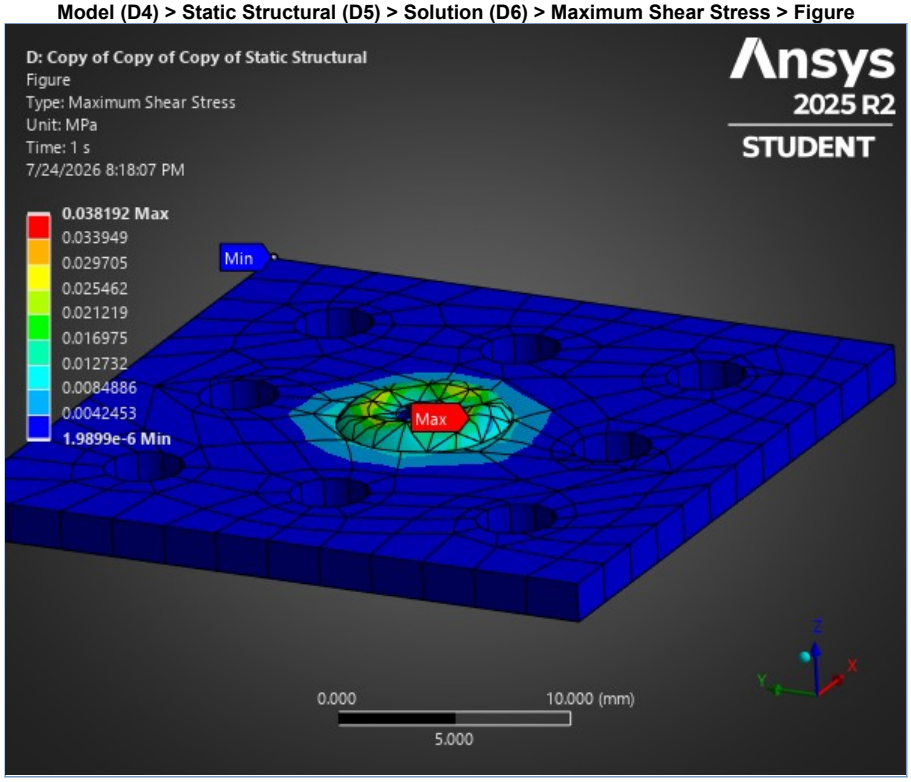


**TABLE 30**
**Model (D4) > Static Structural (D5) > Solution (D6) > Results**

| Object Name | *Stress Intensity* | *Strain Energy* |
|---|---|---|
| State | Solved | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | 2 Bodies | |
| **Definition** | | |
| Type | Stress Intensity | Strain Energy |
| By | Time | |
| Display Time | Last | |
| Separate Data by Entity | No | |
| Calculate Time History | Yes | |
| Identifier | | |
| Suppressed | No | |
| **Integration Point Results** | | |
| Display Option | Averaged | |
| Average Across Bodies | No | |
| **Results** | | |
| Minimum | 3.9798e-006 MPa | 1.3387e-008 mJ |
| Maximum | 7.6384e-002 MPa | 2.3462e-002 mJ |
| Average | 8.3483e-003 MPa | |
| Minimum Occurs On | RUN 4\|Body1 | |
| Maximum Occurs On | RUN 4\|Body9 | RUN 4\|Body1 |
| Total | | 0.52259 mJ |
| **Minimum Value Over Time** | | |
| Minimum | 1.0949e-007 MPa | 1.0789e-011 mJ |
| Maximum | 3.9798e-006 MPa | 1.3387e-008 mJ |
| **Maximum Value Over Time** | | |
| Minimum | 4.3776e-003 MPa | 1.7961e-005 mJ |
| Maximum | 7.6384e-002 MPa | 2.3462e-002 mJ |
| **Information** | | |
| Time | 1. s | |
| Load Step | 1 | |
| Substep | 10 | |
| Iteration Number | 30 | |

**FIGURE 24**
**Model (D4) > Static Structural (D5) > Solution (D6) > Stress Intensity**

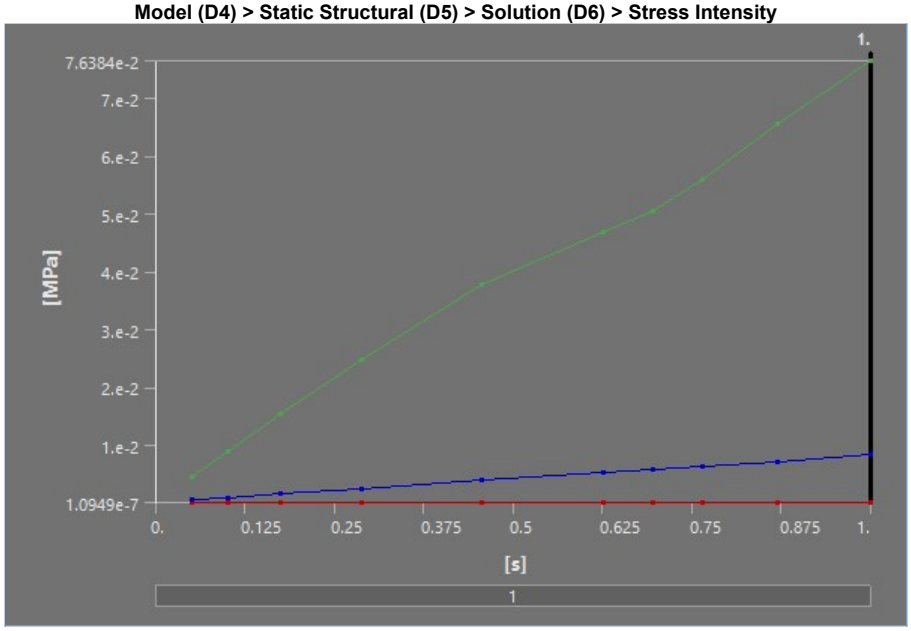



**TABLE 31**
**Model (D4) > Static Structural (D5) > Solution (D6) > Stress Intensity**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 1.0949e-007 | 4.3776e-003 | 4.1703e-004 |
| 0.1 | 2.2051e-007 | 8.7329e-003 | 8.2928e-004 |
| 0.175 | 3.8935e-007 | 1.5211e-002 | 1.4459e-003 |
| 0.2875 | 6.4435e-007 | 2.4769e-002 | 2.3891e-003 |
| | | | |

| | | | |
|---|---|---|---|
| 0.45625 | 1.0635e-006 | 3.7657e-002 | 3.8526e-003 |
| 0.625 | 1.807e-006 | 4.6677e-002 | 5.1036e-003 |
| 0.695 | 2.1778e-006 | 5.0316e-002 | 5.6214e-003 |
| 0.765 | 2.5682e-006 | 5.5961e-002 | 6.1789e-003 |
| 0.87 | 3.1793e-006 | 6.5521e-002 | 7.0869e-003 |
| 1. | 3.9798e-006 | 7.6384e-002 | 8.3483e-003 |

**FIGURE 25**
**Model (D4) > Static Structural (D5) > Solution (D6) > Stress Intensity > Figure**

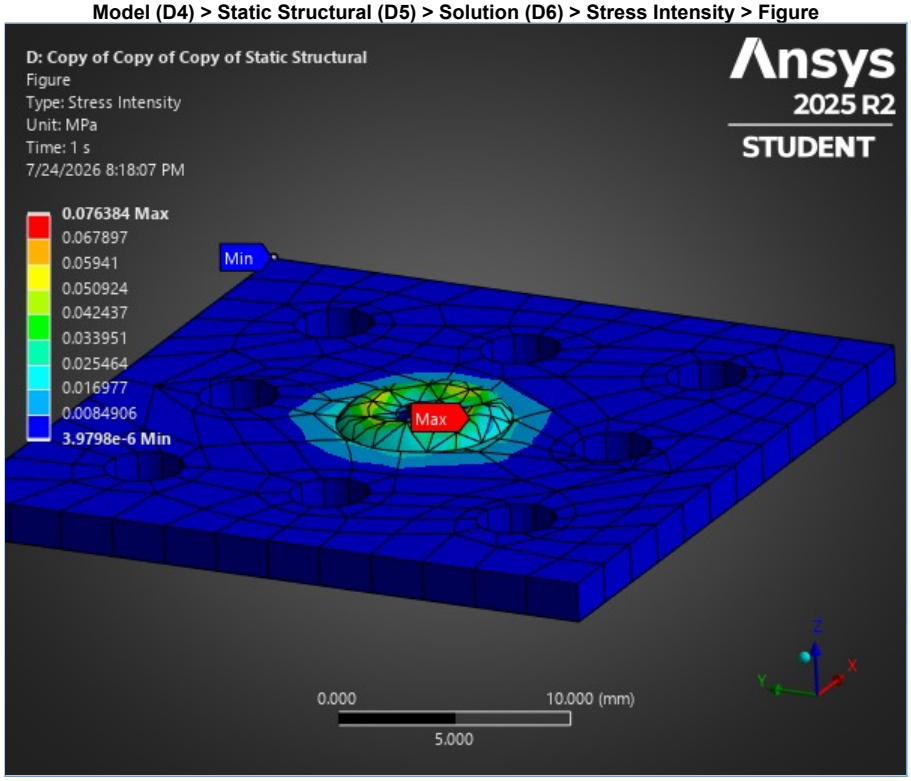



**FIGURE 26**
**Model (D4) > Static Structural (D5) > Solution (D6) > Strain Energy**

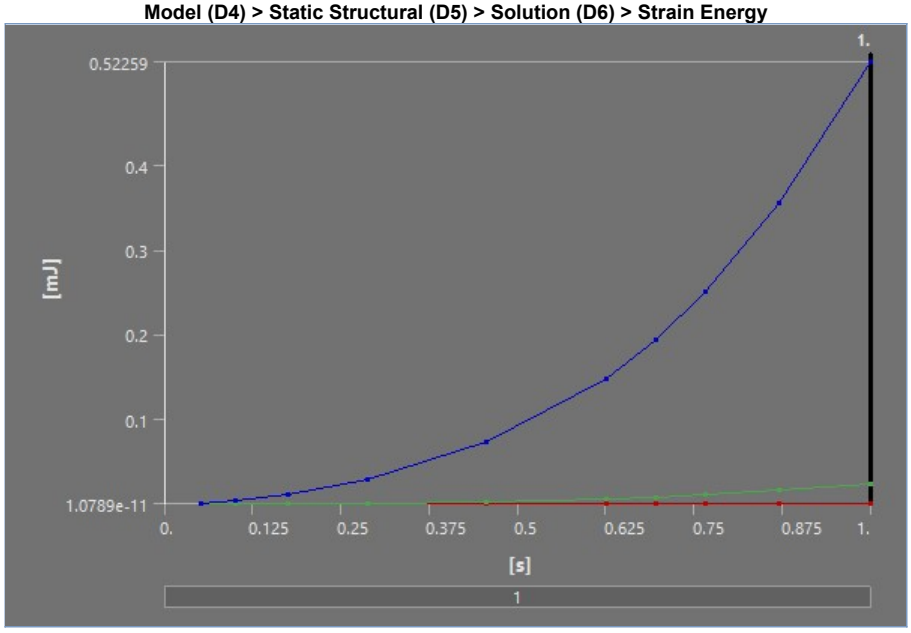



**TABLE 32**
**Model (D4) > Static Structural (D5) > Solution (D6) > Strain Energy**

| Time [s] | Minimum [mJ] | Maximum [mJ] | Total [mJ] |
|---|---|---|---|
| 5.e-002 | 1.0789e-011 | 1.7961e-005 | 8.8617e-004 |
| 0.1 | 4.3888e-011 | 7.1196e-005 | 3.5447e-003 |

| | | | |
|---|---|---|---|
| 0.175 | 1.3744e-010 | 2.1918e-004 | 1.0842e-002 |
| 0.2875 | 3.7953e-010 | 6.0717e-004 | 2.9111e-002 |
| 0.45625 | 1.0387e-009 | 1.5991e-003 | 7.2342e-002 |
| 0.625 | 2.8718e-009 | 4.924e-003 | 0.14674 |
| 0.695 | 4.0989e-009 | 7.1894e-003 | 0.19306 |
| 0.765 | 5.6457e-009 | 9.9911e-003 | 0.24975 |
| 0.87 | 8.5827e-009 | 1.5207e-002 | 0.35505 |
| 1. | 1.3387e-008 | 2.3462e-002 | 0.52259 |

**FIGURE 27**
**Model (D4) > Static Structural (D5) > Solution (D6) > Strain Energy > Figure**

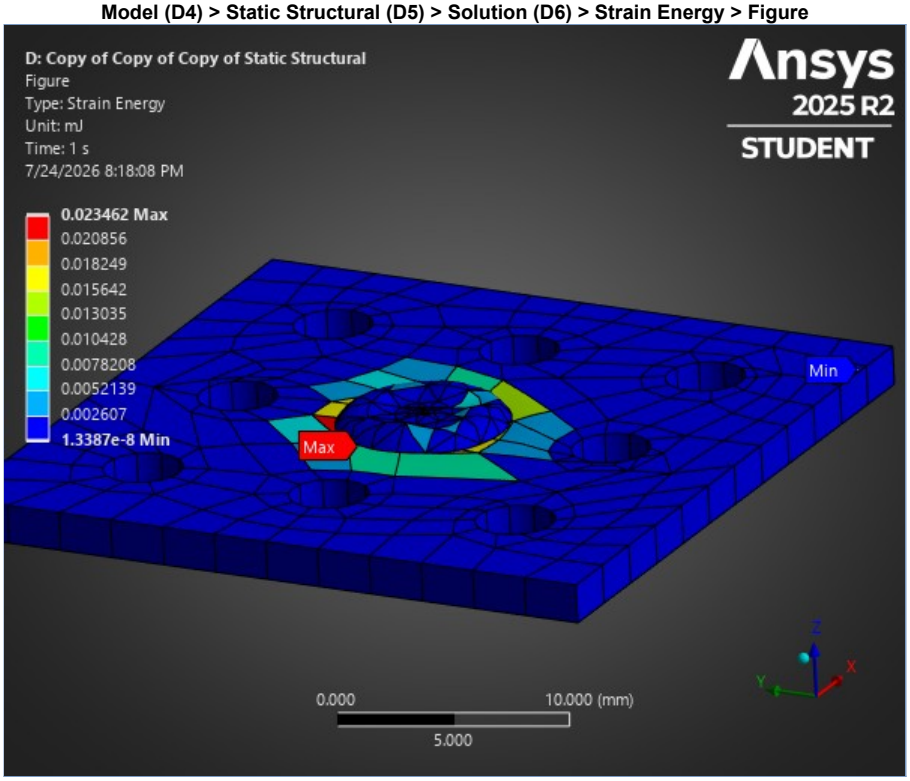



**TABLE 33**
**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tools**

| Object Name | *Contact Tool* |
|---|---|
| State | Solved |
| **Scope** | |
| Scoping Method | Worksheet |

**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tool**

| Name | Contact Side |
|---|---|
| Frictionless - RUN 4\|Body9 To RUN 4\|Body11 | Both |

**TABLE 34**
**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tool > Results**

| Object Name | *Status* |
|---|---|
| State | Solved |
| **Definition** | |
| Type | Status |
| By | Time |
| Display Time | Last |
| Calculate Time History | Yes |
| Identifier | |
| Suppressed | No |
| **Integration Point Results** | |
| Display Option | Averaged |
| **Information** | |
| Time | 1. s |
| Load Step | 1 |
| Substep | 10 |
| Iteration Number | 30 |

**FIGURE 28**
**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tool > Status**

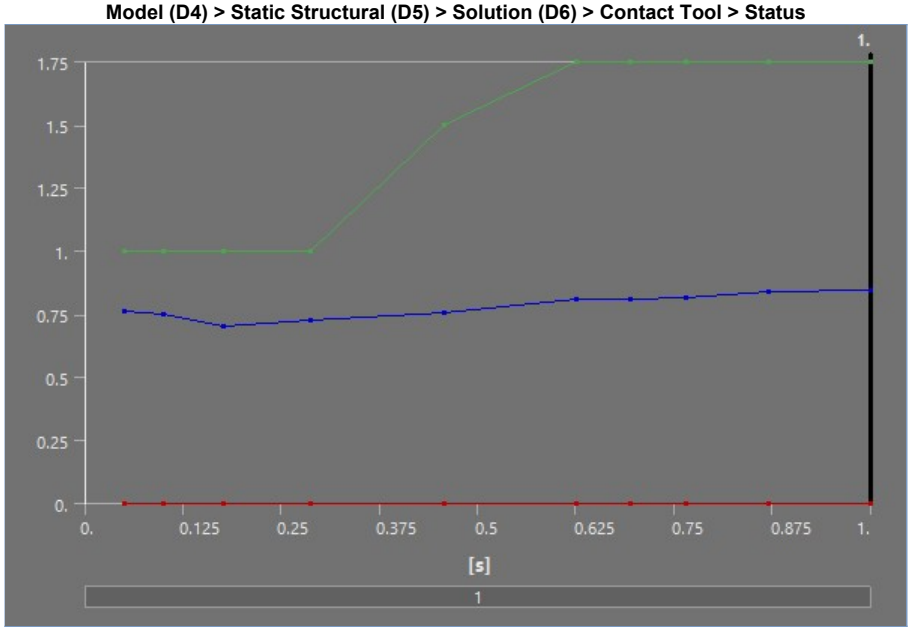



**TABLE 35**
**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tool > Status**

| Time [s] | Minimum | Maximum | Average |
|---|---|---|---|
| 5.e-002 | 0. | 1. | 0.75899 |
| 0.1 | | | 0.75235 |
| 0.175 | | | 0.70011 |
| 0.2875 | | | 0.72399 |
| 0.45625 | | 1.5 | 0.75856 |
| 0.625 | | 1.75 | 0.8075 |
| 0.695 | | | 0.80801 |
| 0.765 | | | 0.81457 |
| 0.87 | | | 0.83997 |
| 1. | | | 0.84365 |

**FIGURE 29**
**Model (D4) > Static Structural (D5) > Solution (D6) > Contact Tool > Status > Figure**

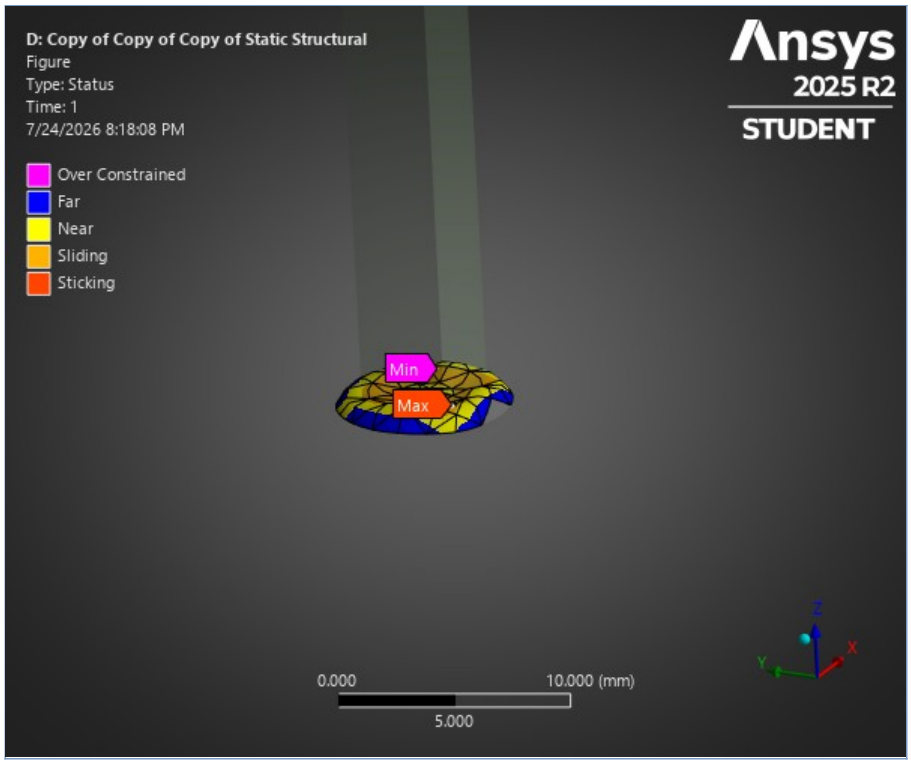



**TABLE 36**
**Model (D4) > Static Structural (D5) > Solution (D6) > Probes**

| Object Name | *Force Reaction* | *Moment Reaction* |
|---|---|---|
| State | Solved | |
| **Definition** | | |
| Type | Force Reaction | Moment Reaction |
| Location Method | Boundary Condition | |
| Boundary Condition | Displacement | |
| Orientation | Global Coordinate System | |
| Suppressed | No | |
| Summation | | Centroid |
| **Options** | | |
| Result Selection | All | |
| Display Time | End Time | |
| **Results** | | |
| X Axis | -7.4968e-003 N | -6.4855e-003 N·mm |
| Y Axis | 1.8146e-005 N | -2.1161e-002 N·mm |
| Z Axis | -1.1661 N | -8.7325e-003 N·mm |
| Total | 1.1662 N | 2.3793e-002 N·mm |
| **Maximum Value Over Time** | | |
| X Axis | 2.3449e-003 N | 4.879e-003 N·mm |
| Y Axis | 1.8146e-005 N | 1.1413e-002 N·mm |
| Z Axis | -2.8407e-002 N | -2.7869e-004 N·mm |
| Total | 1.1662 N | 2.3793e-002 N·mm |
| **Minimum Value Over Time** | | |
| X Axis | -7.4968e-003 N | -1.0259e-002 N·mm |
| Y Axis | -2.4695e-003 N | -2.1161e-002 N·mm |
| Z Axis | -1.1661 N | -8.7325e-003 N·mm |
| Total | 2.8408e-002 N | 1.9785e-003 N·mm |
| **Information** | | |
| Time | 1. s | |
| Load Step | 1 | |
| Substep | 10 | |
| Iteration Number | 30 | |

**FIGURE 30**
**Model (D4) > Static Structural (D5) > Solution (D6) > Force Reaction**

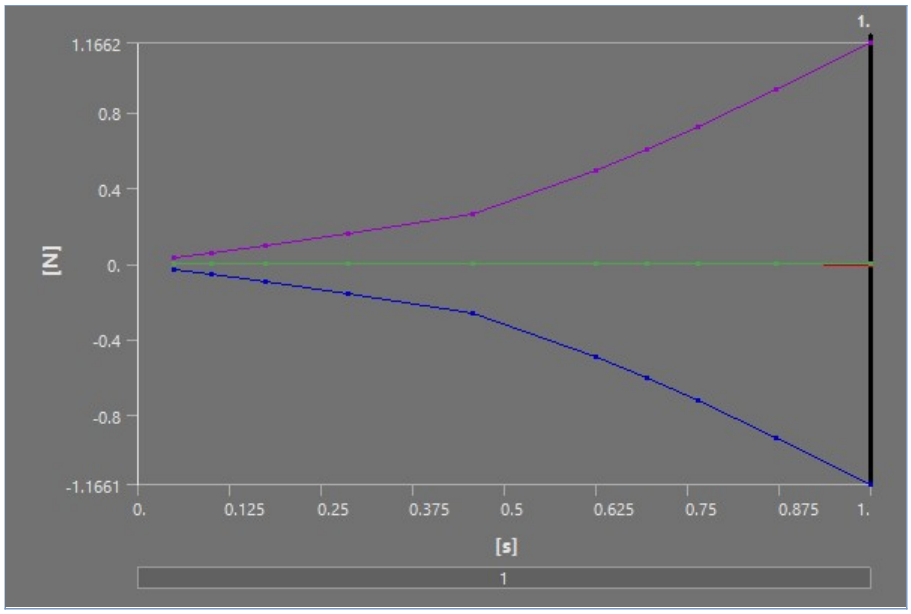



**TABLE 37**
**Model (D4) > Static Structural (D5) > Solution (D6) > Force Reaction**

| Time [s] | Force Reaction (X) [N] | Force Reaction (Y) [N] | Force Reaction (Z) [N] | Force Reaction (Total) [N] |
|---|---|---|---|---|
| 5.e-002 | 1.8712e-004 | -1.1575e-004 | -2.8407e-002 | 2.8408e-002 |
| 0.1 | 3.9085e-004 | -2.5439e-004 | -5.6784e-002 | 5.6786e-002 |
| 0.175 | 7.3546e-004 | -5.2059e-004 | -9.8996e-002 | 9.9e-002 |
| 0.2875 | 1.3224e-003 | -1.0592e-003 | -0.16073 | 0.16074 |
| 0.45625 | 2.3449e-003 | -2.3686e-003 | -0.26309 | 0.26311 |
| 0.625 | 1.482e-003 | -2.4695e-003 | -0.49071 | 0.49072 |
| 0.695 | 2.5037e-004 | -2.1725e-003 | -0.60424 | 0.60424 |
| 0.765 | -8.1059e-004 | -1.6528e-003 | -0.7228 | 0.7228 |
| 0.87 | -3.336e-003 | -9.8692e-004 | -0.91639 | 0.9164 |
| 1. | -7.4968e-003 | 1.8146e-005 | -1.1661 | 1.1662 |

**FIGURE 31**
**Model (D4) > Static Structural (D5) > Solution (D6) > Force Reaction > Figure**

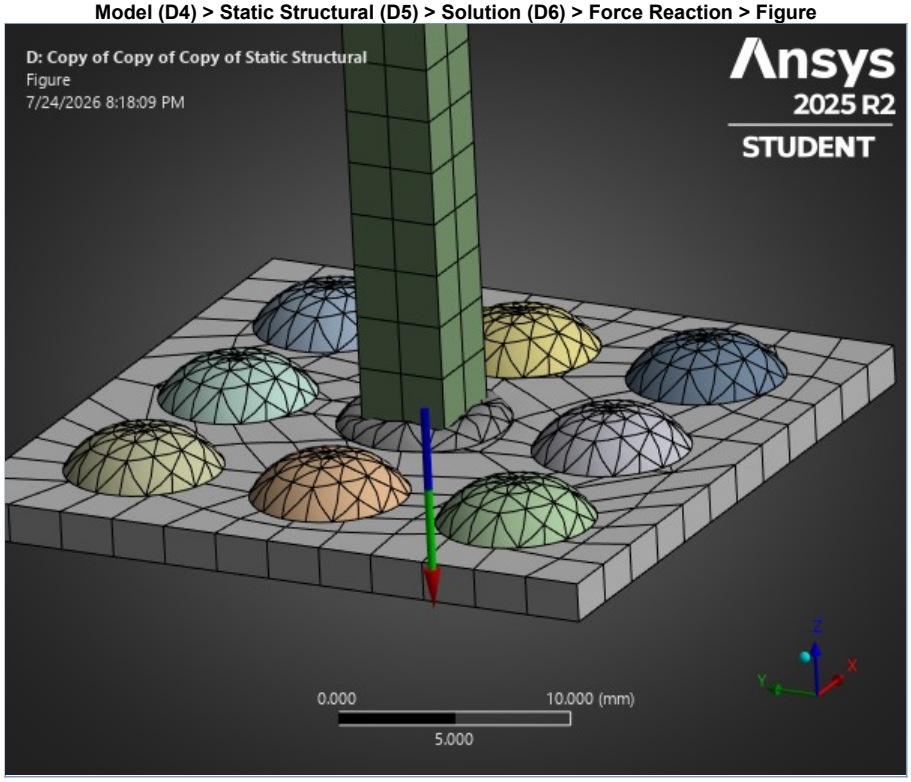


**FIGURE 32**
**Model (D4) > Static Structural (D5) > Solution (D6) > Moment Reaction**

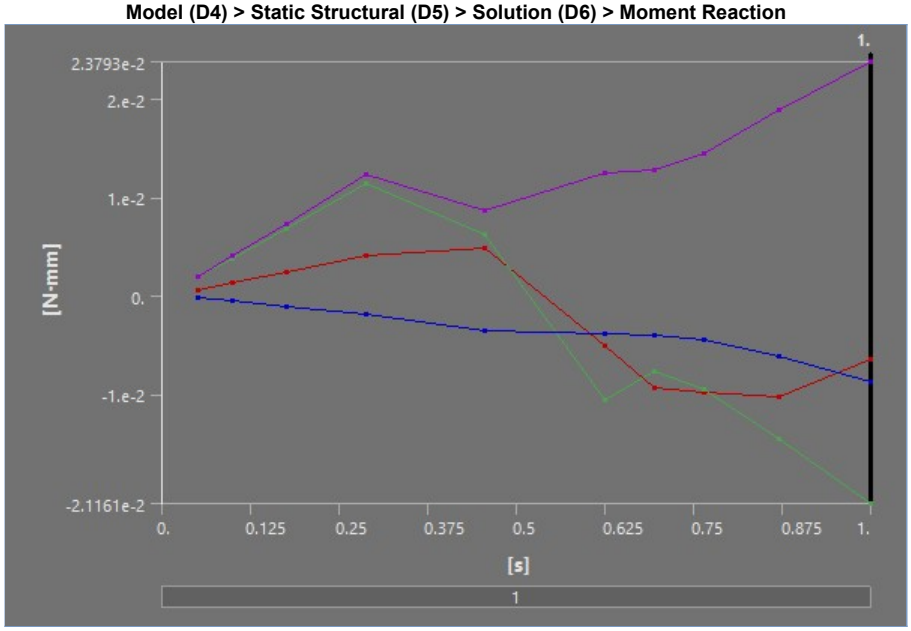
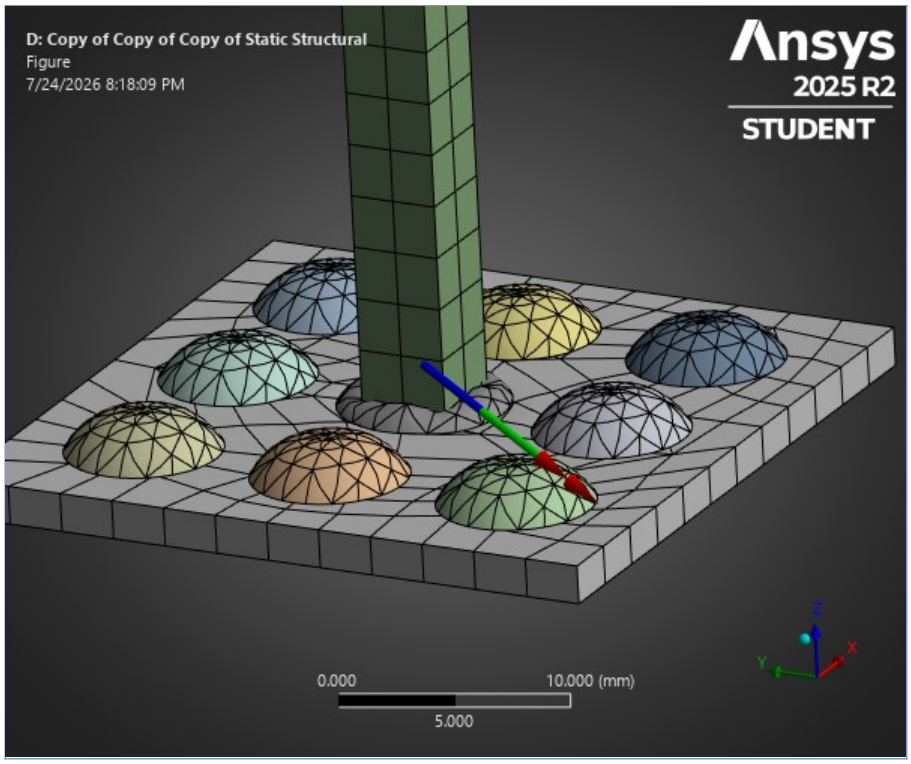



**TABLE 38**
**Model (D4) > Static Structural (D5) > Solution (D6) > Moment Reaction**

| Time [s] | Moment Reaction (X) [N·mm] | Moment Reaction (Y) [N·mm] | Moment Reaction (Z) [N·mm] | Moment Reaction (Total) [N·mm] |
|---|---|---|---|---|
| 5.e-002 | 5.7573e-004 | 1.8723e-003 | -2.7869e-004 | 1.9785e-003 |
| 0.1 | 1.2443e-003 | 3.8087e-003 | -5.8239e-004 | 4.0489e-003 |
| 0.175 | 2.3806e-003 | 6.823e-003 | -1.0841e-003 | 7.3073e-003 |
| 0.2875 | 4.1359e-003 | 1.1413e-002 | -1.9486e-003 | 1.2295e-002 |
| 0.45625 | 4.879e-003 | 6.2247e-003 | -3.5521e-003 | 8.67e-003 |
| 0.625 | -5.031e-003 | -1.0665e-002 | -3.8488e-003 | 1.2405e-002 |
| 0.695 | -9.3321e-003 | -7.765e-003 | -4.1065e-003 | 1.2816e-002 |
| 0.765 | -9.8664e-003 | -9.5369e-003 | -4.4434e-003 | 1.4424e-002 |
| 0.87 | -1.0259e-002 | -1.4537e-002 | -6.1807e-003 | 1.8835e-002 |
| 1. | -6.4855e-003 | -2.1161e-002 | -8.7325e-003 | 2.3793e-002 |

**FIGURE 33**
**Model (D4) > Static Structural (D5) > Solution (D6) > Moment Reaction > Figure**




# Material Data

## *ECOFLEX 00-45*

**TABLE 39**
**ECOFLEX 00-45 > Constants**

| | |
|---|---|
| Density | 1.06e-006 kg mm^-3 |

**TABLE 40**
**ECOFLEX 00-45 > Color**

| Red | Green | Blue |
|---|---|---|
| 181 | 155 | 130 |

**TABLE 41**
**ECOFLEX 00-45 > Neo-Hookean**

| Initial Shear Modulus Mu MPa | Incompressibility Parameter D1 MPa^-1 | Temperature K |
|---|---|---|
| 4.73e-002 | 0.424 | |

## *Plastic, ABS (high-impact) 2*

**TABLE 42**
**Plastic, ABS (high-impact) 2 > Constants**

| | |
|---|---|
| Density | 1.03e-006 kg mm^-3 |
| Tensile Yield Strength | 43.82 MPa |
| Tensile Ultimate Strength | 36.26 MPa |
| Coefficient of Thermal Expansion | 1.84e-004 K^-1 |
| Thermal Conductivity | 1.997e-004 W mm^-1 K^-1 |
| Specific Heat | 1.4e+006 mJ kg^-1 K^-1 |
| Resistivity | 9.95e+016 ohm mm |
| Electric Loss Tangent | 5.916e-003 |
| Relative Permittivity | 2.993 |

**TABLE 43**
**Plastic, ABS (high-impact) 2 > Opacity**

| Red | Green | Blue |
|---|---|---|
| 0 | 153 | 255 |

| Opacity |
|---|
| 0.9 |

| Metallic Finish |
|---|
| |

| | |
|---|---|
| 0 | |

**TABLE 44**
**Plastic, ABS (high-impact) 2 > Isotropic Elasticity**

| Young's Modulus MPa | Poisson's Ratio | Bulk Modulus MPa | Shear Modulus MPa | Temperature K |
|---|---|---|---|---|
| 2405 | 0.4089 | 4399.9 | 853.5 | 296.15 |

**TABLE 45**
**Plastic, ABS (high-impact) 2 > Isotropic Secant Coefficient of Thermal Expansion**

| Zero-Thermal-Strain Reference Temperature K |
|---|
| 296.15 |

**TABLE 46**
**Plastic, ABS (high-impact) 2 > Climate change (CO2-eq)**

| Climate change (CO2-eq) dBA |
|---|
| 3.661 |

**TABLE 47**
**Plastic, ABS (high-impact) 2 > Embodied energy**

| Embodied energy mJ kg^-1 |
|---|
| 9.64e+010 |

**TABLE 48**
**Plastic, ABS (high-impact) 2 > Recycle**

| Recycle |
|---|
| 1 |